

## Full Length Article

## Gaussian processes with prior-model-informed kernel for dynamical system modeling

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## ABSTRACT

Gaussian processes (GPs) are widely used for modeling dynamical systems due to their ability to provide principled uncertainty estimation and to incorporate prior knowledge via the mean or kernel function. However, GPs typically exhibit poor extrapolation performance outside the training region. To address this limitation while preserving the favorable uncertainty behavior of standard GPs, we propose a novel kernel design that integrates prior models—either analytical physical models or imperfect simulators—into the kernel function of the GP. Our approach is motivated by the assumption that similarity in the prior model implies similarity in the true system dynamics. Specifically, we transform the input space using the prior model and apply a base kernel (e.g., Radial Basis Function or Matérn) to construct a prior-model-informed kernel that reflects this assumption. To further enhance modeling flexibility, we add a standard residual kernel to correct discrepancies between the prior model and the true system. This yields our final model: a GP with prior-model-informed kernel (GP-PI-K). Through extensive experiments on benchmark dynamical systems, we demonstrate that GP-PI-K consistently outperforms existing baselines, including standard GPs, GPs with physics-informed or neural network mean functions, and deep kernel learning models, in terms of one-step and multi-step prediction accuracy, uncertainty estimation, and uncertainty calibration. Moreover, GP-PI-K achieves superior performance in downstream applications such as active learning and model-based reinforcement learning.

## 1. Introduction

Probabilistic models have attracted considerable attention due to their ability to provide principled uncertainty estimation (Murphy, 2022). In such models, uncertainty is low in regions covered by training data and high in unexplored areas. This property enables them to distinguish between known (less informative) and unknown (informative) regions in the state space. Consequently, probabilistic models can guide sample-efficient exploration (Ladoss et al., 2022; Pathak et al., 2017; Shyam et al., 2019) and improve model-based control by mitigating the effect of model errors (Deisenroth et al., 2013). These advantages make them particularly effective for modeling dynamical systems in downstream tasks such as model-based reinforcement learning (MBRL) and active learning (AL) (Li et al., 2024).

Among probabilistic models, Gaussian processes (GPs) have been widely adopted for dynamical system modeling, owing to their ability to produce well-calibrated uncertainty estimates (Cuzzolin, 2025; Psaros et al., 2023) and to flexibly incorporate prior knowledge via either the mean or kernel function. One of the most successful MBRL algorithms,

PILCO (Deisenroth et al., 2013), leverages the uncertainty estimates from GPs to reduce the effect of model errors during planning. This leads to significantly improved control performance and extremely high sample efficiency. In the context of actively learning dynamical systems, Buisson-Fenet et al. (2020) and Tang et al. (2022) have utilized GP uncertainty estimates to define information gain metrics, which quantify the informativeness of state-action sequences and enable informative sample acquisition.

Despite these strengths, like many machine learning models, GPs suffer from poor extrapolation capability. Their predictive accuracy is high within the training region but deteriorates rapidly in regions far from the data. To address this issue, incorporating prior knowledge into learning models has emerged as a promising strategy for improving generalization (Fortuin, 2022). For example, physics-informed neural networks (NNs) embed physical laws into NNs to produce more accurate models with fewer training samples (Karniadakis et al., 2021). Similarly, several studies have explored incorporating imperfect physical models into GPs to enhance their extrapolation capability. Tang et al. (2021, 2025) and Riedel and Stulp (2019) first derived analytical physical models

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represented by ordinary differential equations using the Lagrangian method. They then used these models as the mean functions of GPs. Through extensive simulations, they demonstrated that GPs with such physics-informed mean functions (denoted as GP-PI-M) significantly outperform standard zero-mean GPs in terms of extrapolation and predictive accuracy. However, GP-PI-M suffers from two critical limitations: (1) it can yield overconfident uncertainty estimates in practice, and (2) it may suffer from sharp deterioration in long-horizon prediction accuracy when the mean function is imperfect. Specifically, when a physical model is used as the GP's mean function, the learning task becomes modeling the residual between the true system dynamics and the imperfect physical model using a zero-mean GP. This residual modeling often improves extrapolation and one-step prediction accuracy. Nevertheless, the uncertainty behavior of GP-PI-M requires careful interpretation. For fixed kernel hyperparameters, the predictive variance of a GP does not directly depend on the mean function. Therefore, if GP-PI-M exhibits smaller predictive uncertainty in practice, this should not be interpreted as a direct consequence of reduced residual magnitude. A more appropriate interpretation is that introducing an explicit mean function changes the residual-learning problem seen by the GP, which may in turn affect hyperparameter learning and empirically lead to more overconfident uncertainty estimates. Moreover, in the recursive multi-step predictions used in MBRL, errors from the imperfect mean function can accumulate rapidly, thereby degrading long-horizon prediction accuracy. We discuss this issue in more detail later.

Most existing approaches have focused on incorporating physical prior models into the GP's mean function. [Romerres et al. \(2016\)](#) proposed a method to construct a kernel based on a linear physical model, leveraging the equivalence between GPs and Bayesian linear regression. However, their method requires the physical model to be linear in its parameters, which limits its applicability. [Long et al. \(2022\)](#) proposed a general framework, AutoIP, for integrating differential equations into GPs to improve prediction accuracy. AutoIP first samples a set of collocation points that satisfy the given differential equations, and constructs a joint GP prior over the target function and its derivatives via analytical kernel differentiation. It then incorporates physical knowledge using a virtual Gaussian likelihood, which encourages the model to conform to the specified equations. This formulation enables AutoIP to flexibly incorporate arbitrary physical equations without modifying the kernel structure.

Some studies have also explored combining GPs with NNs to leverage their complementary strengths. [Iwata and Ghahramani \(2017\)](#) proposed using an NN as the mean function of the GP (denoted as GP-NN-M). This formulation allows the model to learn a global nonlinear trend from data while relying on the GP to model the residuals and provide uncertainty estimates. [Bayrooti et al. \(2025\)](#) applied GP-NN-M to MBRL, where the NN mean function was responsible for representing complex contact dynamics (i.e., discontinuities). However, similar to GPs with physics-informed mean functions, GP-NN-M can also yield overconfident uncertainty estimates in practice. A more plausible explanation is that introducing an explicit NN mean function changes the residual-learning problem seen by the GP, which may affect hyperparameter learning and lead to poorer uncertainty calibration. To enhance the expressive power of the kernel itself, [Wilson et al. \(2016a,b\)](#) proposed deep kernel learning (denoted as GP-NN-K), in which an NN was used to transform the input space before applying a standard kernel function. This kernel design enables the GP to operate in a learned feature space that captures complex input structures. Their experiments demonstrated that this approach significantly improves predictive accuracy while preserving the uncertainty quantification capability of GPs.

To incorporate prior knowledge into the GP framework while preserving its ability to provide reliable uncertainty estimates, we propose integrating prior models into the kernel function of the GP. For simple systems, analytical physical models derived using the Lagrangian formulation are employed as prior models. For more complex systems, we construct imperfect simulators using physics engines such as Mu-

JoCo ([Todorov et al., 2012](#)) or PyBullet ([Coumans & Bai, 2016](#)) to serve as prior models. Our kernel design is motivated by the assumption that similarity in the prior model implies similarity in the true system function. To incorporate this idea into the GP, we start from a base kernel (i.e., a standard kernel) and transform the input space using the prior model, resulting in a prior-model-informed kernel. When using either the Radial Basis Function kernel or the Matérn kernel as the base kernel, the resulting prior-model-informed kernel reflects this functional-similarity assumption at the level of the GP prior by assigning higher covariance to inputs that are similar in the prior-model space. In this sense, the proposed kernel encodes the functional similarity assumption as an inductive bias, rather than proving that the unknown true system function must satisfy it exactly. To further improve flexibility, we add a standard kernel to correct residual discrepancies between the true system function and its prior model. This yields our final model: a GP with prior-model-informed kernel (GP-PI-K). Owing to its kernel-level integration of prior knowledge, GP-PI-K exhibits strong extrapolation capability while largely preserving the favorable uncertainty behavior of standard GPs. Using both one-step and multi-step prediction accuracy as well as uncertainty estimation and calibration as evaluation metrics, we demonstrate that GP-PI-K consistently outperforms several strong baselines, including GPs with physics-informed means, standard zero-mean GPs, GPs with NN means, and GPs with NN kernels. Additionally, GP-PI-K achieves superior performance in downstream tasks such as AL and MBRL.

## 2. Problem formulation

We consider the problem of learning a discrete-time dynamical system represented by the following transition function:

$$s_{t+1} = f(s_t, a_t), \quad (1)$$

where  $s_t \in \mathbb{R}^S$  denotes the system state at time step  $t$ , and  $a_t \in \mathbb{R}^A$  is the action applied at that step. The system function  $f$  defines the unknown dynamics that govern the system's evolution. By interacting with the system, we can collect a set of  $N$  transitions  $\{(s_j, a_j, s_{j+1})\}_{j=1}^N$ . For each transition, we define the input as  $x_j = (s_j, a_j)$  and the corresponding target as  $y_j = s_{j+1}$ . The training data can thus be compactly expressed as  $X = \{(s_j, a_j)\}_{j=1}^N$  and  $Y = \{s_{j+1}\}_{j=1}^N$ .

### 2.1. Gaussian process modeling

Standard Gaussian processes (GPs) are inherently designed for single-output regression tasks. In this setting, each dimension  $f_i$  of the dynamical system is modeled independently as a GP with a mean function  $m(x)$  and a kernel function  $k(x, x')$  ([Rasmussen & Williams, 2006](#)):

$$f_i(x) \sim \mathcal{GP}(m(x), k(x, x')), \quad i = 1, \dots, S. \quad (2)$$

Given training data  $(X, Y_i)$ , the predictive distribution of  $f_i(x^*)$  at a test input  $x^*$  follows a Gaussian distribution:

$$p(f_i(x^*) | X, Y_i, x^*) = \mathcal{N}(f_i(x^*) | \mu^*, v^*), \quad (3)$$

where the predictive mean  $\mu^*$  and variance  $v^*$  are computed as:

$$\mu^* = m(x^*) + k(x^*, X)[k(X, X) + \sigma_n^2 I]^{-1}(Y_i - m(X)), \quad (4)$$

$$v^* = k(x^*, x^*) - k(x^*, X)[k(X, X) + \sigma_n^2 I]^{-1}k(X, x^*), \quad (5)$$

and where  $\sigma_n^2$  denotes the noise variance, and  $I$  is the identity matrix.

### 2.2. Prior models

To enhance the extrapolation capabilities of GPs, it is common to incorporate a prior model into the GP framework. In this work, we consider two types of prior models, as described below.

### 2.2.1. Physical prior model

For relatively simple dynamical systems, physical models can often be derived from first principles using classical mechanics, such as the Lagrangian method. The resulting physical model is typically formulated as an ordinary differential equation (ODE) involving unknown physical parameters  $\theta_w$ :

$$\dot{s} = g(s, a; \theta_w). \quad (6)$$

Since our focus is on discrete-time dynamical systems, we discretize the physical model using the fourth-order Runge-Kutta (RK4) method:

$$\begin{aligned} k_1 &= g(s_t, a_t; \theta_w), \\ k_2 &= g\left(s_t + \frac{\Delta t}{2} k_1, a_t; \theta_w\right), \\ k_3 &= g\left(s_t + \frac{\Delta t}{2} k_2, a_t; \theta_w\right), \\ k_4 &= g(s_t + \Delta t k_3, a_t; \theta_w), \\ s_{t+1} &= s_t + \frac{\Delta t}{6} (k_1 + 2k_2 + 2k_3 + k_4) := h(s_t, a_t; \theta_w), \end{aligned} \quad (7)$$

where  $\Delta t$  denotes the discretization time step, and  $h(s_t, a_t; \theta_w)$  represents the discretized physical model.

### 2.2.2. Simulator-based prior model

For more complex dynamical systems, deriving ODEs using the Lagrangian method becomes challenging or infeasible. As an alternative, physics engines such as MuJoCo and PyBullet can be used to construct simulators that serve as prior models (Yang et al., 2023). Although these simulators capture the general structure of the real system's dynamics, they often suffer from the following limitations:

- (1) **Inaccurate geometric properties:** The lengths, masses, or dimensions of links in the simulator may deviate from those of the real system.
- (2) **Neglected dynamic effects:** Certain physical effects, such as friction or damping, may be simplified or completely omitted.

We assume that the simulators are fully known. To incorporate such simulators into the GP framework as prior models, we represent the simulator's dynamics using a neural network (NN):

$$y = \text{NN}(x; \theta_{\text{NN}}), \quad (8)$$

where the input  $x$  is the state-action pair  $(s_t, a_t)$  and the output  $y$  is the next state  $s_{t+1}$ . With a set of synthetic transitions  $\{(\tilde{s}_j, \tilde{a}_j, \tilde{s}_{j+1})\}_{j=1}^{N_1}$  generated by the simulator, the NN is trained to mimic the simulator's behavior:

$$\theta_{\text{NN}}^* = \arg \min_{\theta_{\text{NN}}} \sum_{j=1}^{N_1} \left\| \tilde{s}_{j+1} - \text{NN}(\tilde{s}_j, \tilde{a}_j | \theta_{\text{NN}}) \right\|^2. \quad (9)$$

Once trained, the NN is treated as a fixed deterministic function,  $\text{NN}(x; \theta_{\text{NN}}^*)$ .

## 3. Gaussian processes with prior-model-informed kernel

### 3.1. Existing model: GPs with prior-model-informed mean (GP-PI-M)

To improve the extrapolation capability of GPs, a common approach is to incorporate a prior model—either a physical model or a simulator—into the GP's mean function. This method is referred to as GPs with prior-model-informed mean (GP-PI-M) (Riedel & Stulp, 2019; Tang et al., 2025). When a physical prior model is available, GP-PI-M independently models each dimension  $f_i(x)$  of the dynamical system as a GP whose mean function is given by the discretized physical model  $h_i(x; \theta_w)$ :

$$f_i(x) \sim \mathcal{GP}(m_i(x) = h_i(x; \theta_w), k(x, x')), \quad i = 1, \dots, S. \quad (10)$$

The hyperparameters to be optimized include the physical parameters  $\theta_w$ , the kernel parameters  $\theta_k$ , and the noise variance  $\sigma_n^2$ . Training is performed in two steps:

- (1) **Step 1: Optimization of physical parameters.** The physical parameters  $\theta_w$  are optimized by minimizing the squared error between the physical model predictions and the observed outputs:

$$\theta_w^* = \arg \min_{\theta_w} \sum_{j=1}^N \|y_j - h(x_j; \theta_w)\|^2. \quad (11)$$

- (2) **Step 2: Zero-mean GP training on residuals.** After fixing  $\theta_w = \theta_w^*$ , the prior distribution of  $f_i(x)$  becomes a GP with a fixed mean function:

$$f_i(x) \sim \mathcal{GP}(m_i(x) = h_i(x; \theta_w^*), k(x, x')), \quad i = 1, \dots, S. \quad (12)$$

Training a GP with a fixed mean function is equivalent to training a zero-mean GP:

$$f_i(x) - h_i(x; \theta_w^*) \sim \mathcal{GP}(0, k(x, x')), \quad i = 1, \dots, S. \quad (13)$$

The target function becomes the residual between the true system function and the physical prior model, i.e.,  $f_i(x) - h_i(x; \theta_w^*)$ . By training the zero-mean GP on residuals  $\{X, Y_i - h_i(X; \theta_w^*)\}$ , the kernel parameters  $\theta_k$  and the noise variance  $\sigma_n^2$  are optimized.

When a simulator-based prior model is available, GP-PI-M independently models each dimension  $f_i(x)$  as a GP with a fixed mean function  $\text{NN}_i(x; \theta_{\text{NN}}^*)$ :

$$f_i(x) \sim \mathcal{GP}(m_i(x) = \text{NN}_i(x; \theta_{\text{NN}}^*), k(x, x')), \quad i = 1, \dots, S. \quad (14)$$

The hyperparameters to be optimized include the kernel parameters  $\theta_k$  and the noise variance  $\sigma_n^2$ . As in the physical prior model case, a zero-mean GP is trained on residuals  $\{X, Y_i - \text{NN}_i(X; \theta_{\text{NN}}^*)\}$  to optimize hyperparameters:

$$f_i(x) - \text{NN}_i(x; \theta_{\text{NN}}^*) \sim \mathcal{GP}(0, k(x, x')), \quad i = 1, \dots, S. \quad (15)$$

In both cases, the predictive distribution of  $f_i(x^*)$  at a test input  $x^*$  is Gaussian:

$$p(f_i(x^*) | X, Y_i, x^*) = \mathcal{N}(f_i(x^*) | \mu^*, v^*), \quad (16)$$

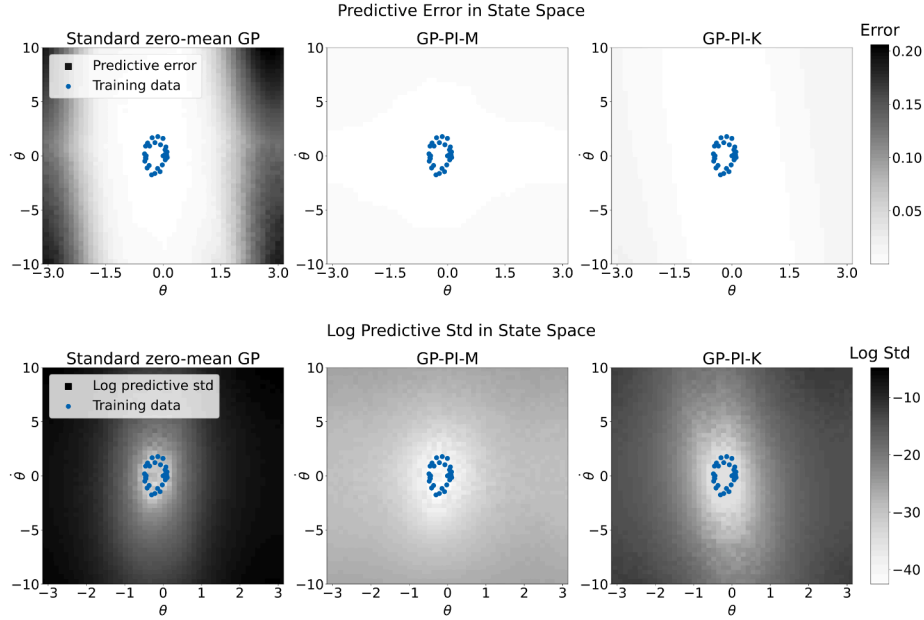
with the predictive mean  $\mu^*$  and variance  $v^*$  given by:

$$\mu^* = m_i(x^*) + k(x^*, X) [k(X, X) + \sigma_n^2 \mathbf{I}]^{-1} (Y_i - m_i(X)), \quad (17)$$

$$v^* = k(x^*, x^*) - k(x^*, X) [k(X, X) + \sigma_n^2 \mathbf{I}]^{-1} k(X, x^*). \quad (18)$$

As shown in Eqs. (13) and (15), GP-PI-M learns a zero-mean GP on the residual between the true system function and the prior model. It should be noted that, for fixed kernel hyperparameters, the predictive variance in Eq. (18) does not directly depend on the mean function. Therefore, if GP-PI-M yields smaller predictive uncertainty in practice, this should not be interpreted as a direct consequence of reduced residual magnitude. A more appropriate interpretation is that introducing an explicit mean function changes the residual-learning problem seen by the GP, which may in turn affect hyperparameter learning and empirically lead to more overconfident uncertainty estimates.

We use a pendulum system as an illustrative example to examine this issue. The system state is  $s = [\theta, \dot{\theta}]^T$ , where  $\theta$  is the angle and  $\dot{\theta}$  is the angular velocity. We train a standard zero-mean GP, GP-PI-M (with a physical prior model as the mean function), and GP-PI-K (introduced in the next subsection) on the same training data, and evaluate their predictive errors and log predictive standard deviations (stds) over the entire state space, as shown in Fig. 1. Larger predictive stds indicate higher uncertainty, and vice versa. In probabilistic modeling, it is generally expected that the model exhibits low uncertainty in regions covered by training data and high uncertainty in regions that are unobserved. The standard zero-mean GP behaves as expected: it yields high accuracy and low uncertainty within the training region, and low accuracy and high uncertainty in unobserved regions. In contrast, GP-PI-M achieves high accuracy even outside the training region due to the incorporation of the physical prior model, but it produces uniformly low uncertainty over a



**Fig. 1.** Comparison of predictive errors and log predictive stds in the pendulum state space  $\{\theta \in [-\pi, \pi], \dot{\theta} \in [-10, 10]\}$ . Top row: predictive errors. Bottom row: log predictive stds. Left: standard zero-mean GP. Middle: GP-PI-M. Right: GP-PI-K. Blue dots denote training data.

large portion of the state space, including unobserved regions. This behavior suggests overconfident uncertainty estimates and is consistent with the calibration results reported later.

Proper uncertainty estimation is crucial for downstream tasks such as active learning and model-based reinforcement learning, where identifying regions of high model uncertainty is essential for effective exploration.

### 3.2. Proposed model: GPs with prior-model-informed kernel (GP-PI-K)

To improve extrapolation while preserving the favorable uncertainty quantification behavior of standard zero-mean GPs, we propose incorporating prior models into the kernel function of the GP, resulting in a novel model, namely GPs with prior-model-informed kernel (GP-PI-K). Unlike GP-PI-M, which incorporates the prior model through the mean function, GP-PI-K incorporates prior knowledge through the covariance structure. Unlike deep kernel learning (GP-NN-K) (Wilson et al., 2016a,b), where the feature transformation is learned purely from data, GP-PI-K uses a structured prior model to define the transformed space. In this sense, GP-PI-K is a prior-model-guided kernel construction method rather than a purely data-driven deep kernel learning method.

GP-PI-K is compatible with both the physical prior model  $h(x; \theta_w)$  and the simulator-based prior model  $NN(x; \theta_{NN}^*)$ . For generality, we denote any prior model as  $\phi(x; \theta_\phi)$ , where:

- $\phi(x; \theta_\phi) = h(x; \theta_w)$  for the physical prior model with trainable parameters,
- $\phi(x; \theta_\phi) = NN(x; \theta_{NN}^*)$  for the simulator-based prior model with fixed parameters.

Since the prior model  $\phi(x; \theta_\phi)$  is an approximation of the true system function  $f(x)$ , we use the following assumption as a motivation for kernel design: **Assumption: Similarity in the prior model implies similarity in the true system function.** For any two inputs  $x$  and  $x'$ , if  $\phi_i(x; \theta_\phi)$  is close to  $\phi_i(x'; \theta_\phi)$ , then  $f_i(x)$  is likely to be close to  $f_i(x')$ :

$$|\phi_i(x; \theta_\phi) - \phi_i(x'; \theta_\phi)| \leq \epsilon_1 \Rightarrow |f_i(x) - f_i(x')| \leq \epsilon_2. \quad (19)$$

Importantly, we do not claim that the unknown true dynamics necessarily satisfy this property exactly. Rather, we use it as an inductive bias to guide kernel construction.

To reflect this motivation in the covariance structure, we start from a base kernel  $k_{\text{base}}(x, x')$  and transform the input space using the prior model  $\phi_i(x; \theta_\phi)$ :

$$k_{\text{base}}(x, x') \longrightarrow k_{\text{base}}(\phi_i(x; \theta_\phi), \phi_i(x'; \theta_\phi)). \quad (20)$$

Reasonable choices for the base kernel include the Radial Basis Function (RBF) kernel and the Matérn kernel:

#### (1) RBF kernel:

$$k_{\text{RBF}}(x, x') = \sigma_f^2 \cdot \exp\left(-\frac{1}{2}(x - x')^T \Lambda^{-1}(x - x')\right), \quad (21)$$

where  $\Lambda := \text{diag}([l_1^2, \dots, l_{S+A}^2])$  are the lengthscales and  $\sigma_f^2$  is the signal variance.

#### (2) Matérn kernel:

$$k_{\text{Matérn}}(x, x') = \sigma_f^2 \cdot \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} d(x, x')\right)^\nu K_\nu\left(\sqrt{2\nu} d(x, x')\right), \quad (22)$$

where  $K_\nu$  is the modified Bessel function of the second kind, and  $d(x, x')$  is the distance with per-dimension lengthscales:

$$d(x, x') = \sqrt{(x - x')^T \Lambda^{-1}(x - x')}, \quad \Lambda := \text{diag}([l_1^2, \dots, l_{S+A}^2]). \quad (23)$$

Here,  $\sigma_f^2$  is the signal variance, and  $\nu > 0$  controls the smoothness of the function: smaller  $\nu$  allows for rougher functions, while  $\nu \rightarrow \infty$  recovers the RBF kernel.

When these kernels are used as the prior-model-informed base kernel in Eq. (25), their inputs become the scalar transformed outputs  $\phi_i(x; \theta_\phi)$  and  $\phi_i(x'; \theta_\phi)$ . Therefore, the corresponding distance reduces to the one-dimensional case and only a scalar lengthscale is used.

Both RBF and Matérn kernels encode the intuition that nearby inputs should have similar function values:

$$|x - x'| \leq \epsilon_3 \Rightarrow |f_i(x) - f_i(x')| \leq \epsilon_4. \quad (24)$$

Motivated by Eq. (19), we define the prior-model-informed kernel as

$$k_{\text{PI}}(x, x') := k_{\text{base}}(\phi_i(x; \theta_\phi), \phi_i(x'; \theta_\phi)). \quad (25)$$

Under this construction, if  $\phi_i(x; \theta_\phi)$  and  $\phi_i(x'; \theta_\phi)$  are close, then the resulting covariance value becomes large due to the property of the RBF or Matérn kernel. Therefore, the proposed kernel reflects the functional-similarity assumption at the level of the GP prior by assigning higher



covariance to inputs that are similar in the prior-model space. In other words, it encodes the assumption in Eq. (19) as an inductive bias, rather than proving that the unknown true system function must satisfy it exactly.

To improve flexibility and robustness to prior mismatch, we further add a standard kernel  $k_{\text{residual}}$  to capture the residual between the true system function and the prior model, and finally model each dimension of the dynamical system independently as a zero-mean GP:

$$f_i(x) \sim \mathcal{GP}(0, k(x, x')), \quad i = 1, \dots, S,$$

$$\text{with } k(x, x') = k_{\text{PI}}(x, x') + k_{\text{residual}}(x, x'). \quad (26)$$

Here, the prior-model-informed kernel  $k_{\text{PI}}(x, x')$  operates on the scalar transformed output  $\phi_i(x; \theta_\phi)$  for the  $i$ -th dimension, and thus uses a scalar lengthscale. Specifically, for the  $i$ -th output dimension, the base-kernel parameters are denoted by  $\theta_k^{\text{base},i} = \{\sigma_{f,\text{base},i}^2, l_{\text{base},i}\}$ , where  $\sigma_{f,\text{base},i}^2$  is the signal variance and  $l_{\text{base},i}$  is the scalar lengthscale in the one-dimensional transformed space  $\phi_i(x; \theta_\phi)$ . By contrast, the residual kernel  $k_{\text{residual}}(x, x')$  is defined on the original input  $x \in \mathbb{R}^{S+A}$  and uses ARD lengthscales. Its parameters are denoted by  $\theta_k^{\text{residual},i} = \{\sigma_{f,\text{res},i}^2, l_{i,1}, \dots, l_{i,S+A}\}$ , where  $\sigma_{f,\text{res},i}^2$  is the signal variance and  $\{l_{i,1}, \dots, l_{i,S+A}\}$  are the ARD lengthscales on the original input space. The additive residual term is important because the prior model is generally imperfect; it allows the GP to preserve the useful similarity structure induced by the prior model while correcting the discrepancy between the prior model and the true dynamics in the original input space.

**Relation to existing GP constructions.** The proposed GP-PI-K can be viewed under a unified feature-kernel formulation. As shown in Eq. (26), for the  $i$ -th output dimension, GP-PI-K applies a base kernel to the prior-model output  $\phi_i(x; \theta_\phi)$  and adds a residual kernel on the original input space. This formulation is related to, but different from, several existing GP constructions.

First, a prior-model feature kernel would directly use prior-model outputs as engineered features, for example  $k_{\text{base}}(\phi_i(x), \phi_i(x'))$ . Such a construction relies entirely on the similarity induced by the prior model. In contrast, GP-PI-K adds a residual kernel on the original input space, which allows the GP to correct discrepancies between the prior model and the true dynamics. Thus, the prior-model-informed kernel provides prior-guided similarity, while the residual kernel preserves flexibility when the prior model is imperfect.

Second, semiparametric GP models typically incorporate prior knowledge through an explicit parametric component, for example by writing  $f_i(x) = m_i(x) + r_i(x)$ , where  $m_i(x)$  is a parametric or prior-model-based mean function and  $r_i(x)$  is a GP residual. GP-PI-K instead keeps a zero-mean GP formulation and incorporates the prior model through the covariance function. Therefore, the prior model affects how inputs are correlated in the GP prior, rather than directly determining the predictive mean.

Third, GP-PI-K is related to deep kernel learning, where a neural network feature map  $q_\eta(x)$  is learned from data and a kernel is applied in the learned feature space, i.e.,  $k_{\text{base}}(q_\eta(x), q_\eta(x'))$ . The key difference is that GP-PI-K uses a structured prior model  $\phi_i(x; \theta_\phi)$  as the feature map. Hence, the transformed space is guided by analytical physical knowledge or a simulator-based prior model, rather than being learned purely from data. This is particularly useful in small-data dynamical-system modeling, where purely data-driven feature learning may be less sample efficient.

Finally, although GP-PI-K uses an additive kernel form, the contribution is not the additivity itself. The contribution lies in constructing an output-dimension-specific prior-model-informed kernel and combining it with a residual kernel so that prior-guided extrapolation and data-driven correction can be achieved within the same GP covariance structure.

All hyperparameters, including the prior model parameters  $\theta_\phi$ , the base and residual kernel parameters  $\{\theta_k^{\text{base}}, \theta_k^{\text{residual}}\}$ , and the noise variance  $\sigma_n^2$ , are jointly learned by maximizing the log marginal likelihood

using the Adam optimizer:

$$\begin{aligned} \ln p(Y_i | X, \theta_\phi, \theta_k^{\text{base}}, \theta_k^{\text{residual}}, \sigma_n^2) \\ = \ln \mathcal{N}(Y_i | 0, k(X, X) + \sigma_n^2 \mathbf{I}) \\ = -\frac{1}{2} Y_i^T [k(X, X) + \sigma_n^2 \mathbf{I}]^{-1} Y_i - \frac{1}{2} \ln |k(X, X) + \sigma_n^2 \mathbf{I}| - \frac{N}{2} \ln(2\pi). \end{aligned} \quad (27)$$

For physical prior models, the prior parameters  $\theta_\phi$  are jointly optimized together with the GP hyperparameters under explicit parameter constraints. In our implementation, the physical parameters are represented by constrained kernel parameters and transformed to satisfy positivity constraints, which restricts optimization to the feasible parameter domain. In addition, a weak regularization term is applied during optimization to discourage unrealistically large deviations of the physical parameters from their nominal values. These constraints and regularization reduce the risk of arbitrarily distorting the prior model to increase the marginal likelihood. The fixed-prior ablation in Section 5.4.1 shows that learning these prior parameters is beneficial, while the residual-kernel ablation shows that the model does not rely solely on the transformed prior-model space. Together, these results indicate that GP-PI-K improves performance through a meaningful combination of prior-guided similarity and residual correction, rather than through unconstrained distortion of the prior model.

We note that our current treatment of the physical prior parameters is still optimization-based rather than fully Bayesian. A more principled but more computationally demanding alternative is to treat the physical prior parameters as random variables, assign explicit prior distributions to them, and infer their posterior distribution jointly with the GP model. This would lead to a hierarchical Bayesian GP model. In this work, we focus on the optimization-based formulation for computational tractability, and we leave the fully Bayesian treatment of physical prior parameters to future work.

Given a test input  $x^*$ , the predictive distribution of  $f_i(x^*)$  is Gaussian:

$$p(f_i(x^*) | X, Y_i, x^*) = \mathcal{N}(f_i(x^*) | \mu^*, v^*), \quad (28)$$

with the predictive mean  $\mu^*$  and variance  $v^*$  given by:

$$\mu^* = k(x^*, X) [k(X, X) + \sigma_n^2 \mathbf{I}]^{-1} Y_i, \quad (29)$$

$$v^* = k(x^*, x^*) - k(x^*, X) [k(X, X) + \sigma_n^2 \mathbf{I}]^{-1} k(X, x^*). \quad (30)$$

When the training dataset is large, the above exact GP inference becomes computationally intractable due to its  $\mathcal{O}(N^3)$  computational complexity, where  $N$  is the number of training points. To address this, we adopt the stochastic variational Gaussian process (SVGP) framework, which introduces a set of inducing points and optimizes a variational lower bound on the marginal likelihood (Gardner et al., 2018; Hensman et al., 2015). Training is performed efficiently via mini-batch stochastic optimization, enabling scalability to large datasets. All model parameters are jointly optimized by maximizing the evidence lower bound.

As illustrated in Fig. 1, GP-PI-K achieves high predictive accuracy while maintaining informative uncertainty in regions far from the training data. This behavior is consistent with our goal of improving predictive performance while largely preserving the favorable uncertainty behavior of standard zero-mean GPs.

Finally, as in the other GP baselines, each output dimension is modeled independently. This design improves scalability and implementation simplicity, but it ignores cross-output correlations, which may affect joint calibration in multi-step rollouts and downstream MBRL. We leave structured multi-output extensions to future work.

## 4. Experiments

### 4.1. Dynamical systems

We select four representative continuous control environments from MuJoCo (Todorov et al., 2012) as the ground-truth dynamical systems, as shown in Fig. 2. For each system, we define a corresponding prior

model—either a physical prior model or a simulator-based prior model—used in the proposed GP-PI framework.

#### (1) 2-link Reacher

This is a two-link robot arm with state  $s = [\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2]^T$ , where  $\theta_1, \theta_2$  are joint angles and  $\dot{\theta}_1, \dot{\theta}_2$  are angular velocities. The action is a 2-dimensional torque vector  $a = [a_1, a_2]^T$ . To handle angular periodicity, we augment the state as  $s_{\text{aug}} = [\cos \theta_1, \cos \theta_2, \sin \theta_1, \sin \theta_2, \dot{\theta}_1, \dot{\theta}_2]^T$ . The physical model (i.e., ODE) is derived via the Lagrangian formulation:

$$\begin{bmatrix} \frac{1}{3}m_1l_1^2 + \frac{1}{3}m_2l_2^2 + m_2l_1^2 + m_2l_1l_2 \cos \theta_2 & \frac{1}{3}m_2l_2^2 + \frac{1}{2}m_2l_1l_2 \cos \theta_2 \\ \frac{1}{3}m_2l_2^2 + \frac{1}{2}m_2l_1l_2 \cos \theta_2 & \frac{1}{3}m_2l_2^2 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} = C, \quad (31)$$

where

$$C = \begin{bmatrix} m_2l_1l_2 \sin \theta_2 \dot{\theta}_1 \dot{\theta}_2 + \frac{1}{2}m_2l_1l_2 \sin \theta_2 \dot{\theta}_2^2 + a_1 \\ -\frac{1}{2}\dot{\theta}_1^2 m_2l_1l_2 \sin \theta_2 + a_2 \end{bmatrix}. \quad (32)$$

From Eqs. (31) and (32), we can obtain the ODE and then discretize it to obtain the physical prior model:

$$\dot{s} = g(s, a; \theta_w) \xrightarrow{\text{Discretizing}} s_{t+1} = h(s_t, a_t; \theta_w), \quad (33)$$

where the unknown physical parameters  $\theta_w = \{m_1, m_2, l_1, l_2\}$  include masses and lengths of links. Note that this prior model is only approximate, as it neglects friction and damping.

#### (2) 3-link Reacher

This is a three-link robot arm with state  $s = [\theta_1, \theta_2, \theta_3, \dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3]^T$ , where  $\theta_1, \theta_2, \theta_3$  are joint angles and  $\dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3$  are angular velocities. The action is a 3-dimensional torque vector  $a = [a_1, a_2, a_3]^T$ . We augment the state as  $s_{\text{aug}} = [\cos \theta_1, \cos \theta_2, \cos \theta_3, \sin \theta_1, \sin \theta_2, \sin \theta_3, \dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3]^T$ . The physical model can be derived via the Lagrangian formulation:

$$M(q)\ddot{q} + C(q, \dot{q}) = a, \quad (34)$$

where  $q = [q_1, q_2, q_3]^T = [\theta_1, \theta_2, -\theta_3]^T$  and  $M(q)$  is the inertial matrix:

$$M(q) = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{21} & M_{22} & M_{23} \\ M_{31} & M_{32} & M_{33} \end{bmatrix}, \quad (35)$$

with entries:

$$\begin{cases} M_{11} = e_1 + e_2 + e_4 + 2e_3 \cos q_2 + 2e_5 \cos(q_2 + q_3) + 2e_6 \cos q_3, \\ M_{12} = e_2 + e_4 + e_3 \cos q_2 + e_5 \cos(q_2 + q_3) + 2e_6 \cos q_3, \\ M_{13} = e_4 + e_5 \cos(q_2 + q_3) + e_6 \cos q_3, \\ M_{22} = e_2 + e_4 + 2e_6 \cos q_3, \\ M_{23} = e_4 + e_6 \cos q_3, \\ M_{33} = e_4, \\ M_{21} = M_{12}, \\ M_{31} = M_{13}, \\ M_{32} = M_{23}. \end{cases} \quad (36)$$

The constants  $\{e_1, \dots, e_6\}$  depend on the unknown physical parameters  $\theta_w = \{m_1, m_2, m_3, l_1, l_2, l_3\}$ , where  $m_i$  and  $l_i$  denote the mass and length of each link:

$$\begin{cases} e_1 = \frac{7}{12}m_1l_1^2 + (m_2 + m_3)l_1^2, \\ e_2 = \frac{1}{3}m_2l_2^2 + m_3l_2^2, \\ e_3 = \frac{1}{2}m_2l_1l_2 + m_3l_1l_2, \\ e_4 = \frac{1}{3}m_3l_3^2, \\ e_5 = \frac{1}{2}m_3l_1l_3, \\ e_6 = \frac{1}{2}m_3l_2l_3. \end{cases} \quad (37)$$

The Coriolis and centrifugal vector  $C(q, \dot{q})$  is:

$$C(q, \dot{q}) = \begin{bmatrix} -e_3(2\dot{q}_1 + \dot{q}_2)\dot{q}_2 \sin q_2 - e_6(2\dot{q}_1 + 2\dot{q}_2 + \dot{q}_3)\dot{q}_3 \sin q_3 \\ -e_5(2\dot{q}_1 + \dot{q}_2 + \dot{q}_3)(\dot{q}_2 + \dot{q}_3) \sin(q_2 + q_3), \\ e_3\dot{q}_1^2 \sin q_2 + e_5\dot{q}_1^2 \sin(q_2 + q_3) - e_6(2\dot{q}_1 + 2\dot{q}_2 + \dot{q}_3)\dot{q}_3 \sin q_3, \\ e_5\dot{q}_1^2 \sin(q_2 + q_3) + e_6(\dot{q}_1 + \dot{q}_2)^2 \sin q_3 \end{bmatrix}. \quad (38)$$

Given Eq. (34), we can solve for  $\ddot{q} = M(q)^{-1}(a - C(q, \dot{q}))$  to obtain the ODE and the physical prior model:

$$\dot{s} = g(s, a; \theta_w) \xrightarrow{\text{Discretizing}} s_{t+1} = h(s_t, a_t; \theta_w). \quad (39)$$

The prior model is imperfect because friction and damping are not considered.

#### (3) Half-Cheetah

This is a 2-dimensional robot with 6 actuated joints. The state is defined as:

$$s = \left[ \frac{x_{t+1} - x_t}{dt}, z, \theta_1, \dots, \theta_7, \dot{x}, \dot{z}, \dot{\theta}_1, \dots, \dot{\theta}_7 \right]^T, \quad (40)$$

where  $x$  and  $z$  denote the horizontal and vertical coordinates of the front tip,  $\{\theta_i\}_{i=1}^7$  are joint angles, and  $\dot{x}, \dot{z}, \{\dot{\theta}_i\}_{i=1}^7$  are the corresponding velocities. The term  $\frac{x_{t+1} - x_t}{dt}$  is included to estimate the average forward velocity, which is required for reward computation in model-based control. To capture angular periodicity, we augment the state as:

$$s_{\text{aug}} = \left[ \frac{x_{t+1} - x_t}{dt}, z, \cos \theta_1, \dots, \cos \theta_7, \sin \theta_1, \dots, \sin \theta_7, \dot{x}, \dot{z}, \dot{\theta}_1, \dots, \dot{\theta}_7 \right]^T. \quad (41)$$

For complex dynamical systems such as Half-Cheetah, deriving the ODE via Lagrangian mechanics is challenging due to the high-dimensional state and the presence of contact dynamics. Therefore, we construct a simulator as the prior model (Miller et al., 2024; Niu et al., 2025), as illustrated in Fig. 2. The simulator is intentionally made imperfect to simulate real-world modeling discrepancies. Specifically, it uses a torso that is 1.5× longer and assumes only half the friction and damping compared to the true system. We then generate synthetic state transitions from the simulator and represent its dynamics as an NN with fixed parameters, which is used as a simulator-based prior model.

#### (4) 7-DOF Reacher

This is a 3-dimensional robot arm with 7 actuated joints. The state is defined as:

$$s = [\theta_1, \dots, \theta_7, \dot{\theta}_1, \dots, \dot{\theta}_7]^T, \quad (42)$$

where  $\{\theta_i\}_{i=1}^7$  are joint angles, and  $\{\dot{\theta}_i\}_{i=1}^7$  are angular velocities. To address angular discontinuities, we augment the state as:

$$s_{\text{aug}} = [\cos \theta_1, \dots, \cos \theta_7, \sin \theta_1, \dots, \sin \theta_7, \dot{\theta}_1, \dots, \dot{\theta}_7]^T. \quad (43)$$

Due to the difficulty of deriving analytical dynamics for high-DOF arms, we construct a simulator as the prior model. To simulate real-world model mismatch, the simulator is deliberately made imperfect by extending the upper-arm and forearm by 1.5× and reducing both friction and damping to half of the true values. Using synthetic state transitions from the simulator, we train an NN as the simulator-based prior model.

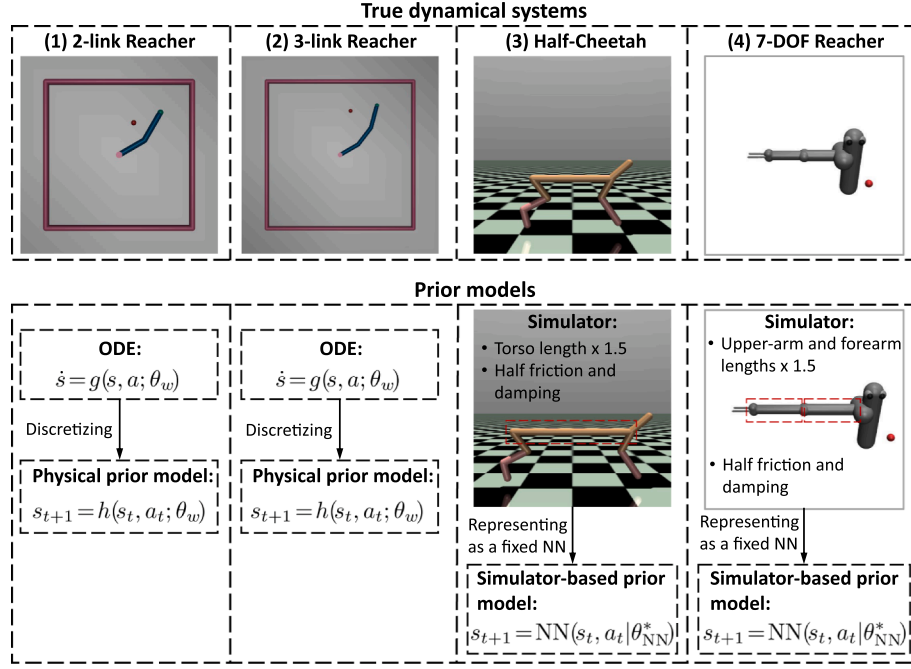
### 4.2. Baselines

We compare the proposed model, GPs with prior-model-informed kernel (GP-PI-K), against the following seven baselines:

(1) **Standard zero-mean GPs (GP)**. Each dimension of the dynamical system is modeled independently using a zero-mean GP with a standard kernel:

$$f_i(x) \sim \mathcal{GP}(m(x) = 0, k(x, x')), \quad i = 1, \dots, S. \quad (44)$$

The hyperparameters to be optimized include the kernel parameters and the noise variance.



**Fig. 2.** Four dynamical systems and their corresponding prior models. For 2-link Reacher and 3-link Reacher, the prior model is an ODE derived from the physical system. For Half-Cheetah and 7-DOF Reacher, the prior model is a simulator with deliberately introduced inaccuracies, which is then represented as an NN with fixed parameters.

- (2) **GPs with NN mean (GP-NN-M)** (Iwata & Ghahramani, 2017). Each dimension is modeled independently as a GP with an NN mean function and a standard kernel:

$$f_i(x) \sim \mathcal{GP}(m(x) = \text{NN}(x), k(x, x')), \quad i = 1, \dots, S. \quad (45)$$

The trainable parameters include the NN parameters, the kernel parameters, and the noise variance.

- (3) **GPs with NN kernel (GP-NN-K)** (Wilson et al., 2016a,b). Each dimension is modeled independently as a zero-mean GP with an NN-based kernel:

$$f_i(x) \sim \mathcal{GP}(m(x) = 0, k_{\text{base}}(\text{NN}(x), \text{NN}(x'))), \quad i = 1, \dots, S, \quad (46)$$

where the input space of the kernel is transformed via an NN, and the base kernel  $k_{\text{base}}$  is a standard kernel. The learnable parameters include the NN parameters, the base kernel parameters, and the noise variance.

- (4) **GPs with prior-model-informed mean (GP-PI-M)** (Riedel & Stulp, 2019; Tang et al., 2025). Each dimension is modeled independently as a GP with a prior model  $\phi(x; \theta_\phi)$  as the mean function and a standard kernel:

$$f_i(x) \sim \mathcal{GP}(m(x) = \phi_i(x; \theta_\phi), k(x, x')), \quad i = 1, \dots, S. \quad (47)$$

For physical prior models, the learnable parameters include the physical parameters, the kernel parameters and the noise variance. For simulator-based prior models, only the kernel parameters and the noise variance are optimized.

- (5) **Prior model (PM)**. We also directly evaluate the performance of the prior model itself. This serves to illustrate the modeling error of the prior and to validate the necessity of residual learning through GPs.

- (6) **AutoIP** (Long et al., 2022). AutoIP is a physics-informed GP framework originally designed for continuous-time dynamical systems. To adapt AutoIP to our discrete-time prediction task without introducing errors from first-order finite-difference derivative reconstruction, we construct continuous-time training targets directly from the ground-truth system. Specifically, for each state-action pair  $(s_t, a_t)$ , we obtain the corresponding continuous-time state deriva-

tive  $\dot{s}_t$  and train AutoIP to model the continuous-time vector field:

$$\dot{s}_t = g_{\text{AutoIP}}(s_t, a_t). \quad (48)$$

AutoIP then incorporates the given ODE as physical prior knowledge under its original continuous-time formulation. At prediction time, the learned continuous-time vector field is mapped to the next-state prediction using the same RK4 discretization described in Eq. (7):

$$\hat{s}_{t+1} = \text{RK4}(s_t, a_t, \hat{g}_{\text{AutoIP}}, \Delta t), \quad (49)$$

where  $\hat{g}_{\text{AutoIP}}$  denotes the learned continuous-time vector field. This adaptation avoids approximating the derivative by  $(s_{t+1} - s_t)/\Delta t$  and therefore provides a fairer comparison for AutoIP under its original continuous-time modeling framework. Since AutoIP requires explicit differential equations as prior knowledge, it is only applicable to the 2-link Reacher and 3-link Reacher systems in our experiments, for which ODE-based prior models are available.

- (7) **Deep Ensembles (DE)** (Lakshminarayanan et al., 2017). Deep Ensembles is a strong uncertainty-aware NN baseline. We train an ensemble of 10 independently initialized NNs, each of which predicts the next-state output. The ensemble mean is used as the predictive mean, while the ensemble variance is used to quantify predictive uncertainty. This baseline provides a representative non-GP probabilistic modeling approach for comparison.

#### 4.3. Evaluation metrics

We compare the proposed model with baselines using the following metrics. Unless otherwise specified, all reported results are averaged over 20 random seeds and are presented as mean  $\pm$  95% confidence interval (CI).

##### (1) Model prediction performance

- (a) **One-step prediction accuracy**. The test set consists of transitions  $\{s_i, a_i, s_{i+1}\}_{i=1}^{M_1}$  collected from: 1) applying a model-free reinforcement learning (RL) algorithm, 2) the optimal policy, and 3) random actions. For each test transition  $(s_i, a_i, s_{i+1})$ , the predictive mean is denoted as  $\mu_{i+1}$ . The root mean square error (RMSE)

for one-step prediction in the  $k$ -th state dimension is given by:

$$\text{RMSE}_{\text{one}}[k] := \sqrt{\frac{1}{M_1} \sum_{i=1}^{M_1} (\mu_{i+1}[k] - s_{i+1}[k])^2}, \quad (50)$$

where  $s_{i+1}[k]$  is the  $k$ -th dimension of the state.

- (b) **Multi-step prediction accuracy.** A total of  $M_2$  test trajectories are collected using the same data sources as above. From each trajectory,  $M_3$  states are randomly selected as the initial states for multi-step prediction. For instance, given an initial state  $s_i$  and a future action sequence  $\{a_i, \dots, a_{i+L-1}\}$ , the model recursively predicts the state trajectory  $\{\hat{s}_{i+1}, \dots, \hat{s}_{i+L}\}$ :

$$s_i \xrightarrow{a_i} \hat{s}_{i+1} \sim p(\hat{s}_{i+1}|s_i, a_i) \xrightarrow{a_{i+1}} \hat{s}_{i+2} \sim p(\hat{s}_{i+2}|\hat{s}_{i+1}, a_{i+1}) \rightarrow \dots \quad (51)$$

At each step, the next state is sampled from its predictive distribution. Denote all the predicted states as  $\{\hat{s}_j\}_{j=1}^{M_2 M_3}$  and the corresponding ground-truth states as  $\{s_j\}_{j=1}^{M_2 M_3}$ . For each state dimension  $k$ , the  $L$ -step RMSE is computed as:

$$\text{RMSE}_{L\text{-step}}[k] := \sqrt{\frac{1}{M_2 M_3} \sum_{j=1}^{M_2 M_3} (\hat{s}_j[k] - s_j[k])^2}. \quad (52)$$

To account for the varying magnitudes across state dimensions, we normalize the RMSE by the range of each state dimension and compute the average over all dimensions:

$$\text{RMSE}_{L\text{-step}}^{\text{Norm}} := \text{Mean} \left( \frac{\text{RMSE}_{L\text{-step}}}{\max(s) - \min(s)} \right), \quad (53)$$

where  $\max(s)$  and  $\min(s)$  denote the maximum and minimum values of each state dimension over the test dataset, respectively.

- (c) **Uncertainty estimation and calibration.** We evaluate uncertainty quality from both density estimation and calibration perspectives.

First, we use the mean negative log predictive density (MNLP), which is commonly adopted to assess the quality of predictive uncertainty. A smaller MNLP value indicates better uncertainty estimation. For each test transition  $(s_i, a_i, s_{i+1})$ , the model outputs the predictive distribution of the next state  $\hat{s}_{i+1} \sim p(\hat{s}_{i+1}|s_i, a_i)$ . The MNLP for the  $k$ -th state dimension is defined as:

$$\text{MNLP}[k] := \frac{1}{M_1} \sum_{i=1}^{M_1} -\ln p(\hat{s}_{i+1}[k]|s_i, a_i) \Big|_{\hat{s}_{i+1}[k]=s_{i+1}[k]}, \quad (54)$$

where  $p(\hat{s}_{i+1}[k]|s_i, a_i) \Big|_{\hat{s}_{i+1}[k]=s_{i+1}[k]}$  denotes the value of the predictive density evaluated at the ground-truth value.

Second, we evaluate uncertainty calibration using **empirical coverage**. Specifically, for each predictive interval with nominal coverage level  $\alpha$ , we compute the fraction of test targets that fall inside the corresponding predictive interval. We report the empirical coverages at the 68% and 95% nominal levels, denoted as cov68 and cov95, respectively. Values closer to the nominal levels indicate better-calibrated uncertainty.

Third, we use **reliability diagrams** to visualize calibration quality over a range of nominal coverage levels. A reliability diagram plots the empirical coverage against the nominal coverage. Better-calibrated uncertainty is indicated by curves closer to the diagonal line. We also summarize each reliability diagram using the **reliability error** (RE), which measures the average absolute deviation between empirical and nominal coverages over the evaluated nominal coverage levels. A smaller RE indicates better overall calibration.

Finally, we additionally report the **continuous ranked probability score** (CRPS) and the **interval score** to evaluate the overall quality of the predictive distribution. CRPS is a proper scoring rule that jointly reflects predictive accuracy and uncertainty calibration. The interval score evaluates both the width and the coverage of predictive intervals, thereby measuring the trade-off

between sharpness and calibration. We report interval scores at the 68% and 95% nominal levels, denoted as IS68 and IS95, respectively. Lower CRPS and interval scores indicate better probabilistic prediction quality.

- (2) **Active learning (AL) of dynamical systems.** AL iteratively selects the most informative action sequence by maximizing the information gain  $\text{IG}(a_0, \dots, a_{T'-1})$  over a horizon  $T'$ :

$$\{a_0^*, \dots, a_{T'-1}^*\} = \arg \max_{\{a_0, \dots, a_{T'-1}\}} \text{IG}(a_0, \dots, a_{T'-1}),$$

$$\text{subject to } s_{i+1} \sim p(s_{i+1}|s_i, a_i, X, Y). \quad (55)$$

A commonly used formulation of information gain is the accumulated expected differential entropy of the predictive distribution:

$$\text{IG}(a_0, \dots, a_{T'-1}) := \sum_{i=0}^{T'-1} \mathbb{E}_{s_i} [\mathcal{H}[s_{i+1}]], \quad (56)$$

where  $\mathcal{H}[\cdot]$  denotes the differential entropy. Since information gain is directly determined by the model's predictive uncertainty, reliable uncertainty estimation is critical for effective active learning. Underestimating uncertainty can lead to the selection of less informative samples, thereby degrading sample efficiency and slowing learning progress.

- (3) **Model-based reinforcement learning (MBRL).** In MBRL, the agent leverages a learned dynamics model to simulate environment interactions for planning. Reliable uncertainty estimation is essential for both exploration and planning. During exploration, uncertainty guides the agent toward uncertain regions of the state space to accelerate knowledge acquisition. During planning, uncertainty-aware trajectories can reduce the risk of overconfident predictions that lead to suboptimal behavior. In particular, underestimating uncertainty can cause the agent to overexploit inaccurate model predictions, resulting in poor sample efficiency and degraded control performance.

## 5. Results and discussion

For the 2-link Reacher and 3-link Reacher systems, we adopt the RBF kernel for all GP-based models. For the Half-Cheetah and 7-DOF Reacher systems, we employ the Matérn kernel. All experiments are repeated with 20 different random seeds. We begin by evaluating the models' prediction performance, including one-step prediction accuracy, uncertainty estimation and calibration, and multi-step prediction accuracy. We then assess their performance in downstream tasks, including active learning (AL) and model-based reinforcement learning (MBRL).

### 5.1. Model prediction performance

**Table 1** summarizes the number of training and test data points used for each system.

#### 5.1.1. One-step prediction accuracy

**Tables 2–5** summarize the RMSEs for one-step prediction across all systems. Each row corresponds to one dimension of the state, while each column denotes a different model. The reported values represent the means and 95% confidence intervals (CIs) over 20 random seeds. The overall trends can be summarized as follows:

- 2-link Reacher and 3-link Reacher (simple systems):

$$\text{GP-PI-K} \approx \text{GP-PI-M} > \text{AutoIP} > \text{GP} > \text{GP-NN-M} \approx \text{GP-NN-K} > \text{PM, DE}, \quad (57)$$

- Half-Cheetah and 7-DOF Reacher (complex systems):

$$\text{GP-PI-K} \approx \text{GP-PI-M} > \text{GP-NN-M} \approx \text{GP-NN-K} > \text{GP} > \text{PM, DE}. \quad (58)$$

Here, the symbol “>” indicates better performance (i.e., lower RMSE), and “≈” denotes comparable performance. The results of one-step prediction accuracy can be interpreted as follows:



**Table 1**  
Numbers of training and test data.

|                             | 2-link Reacher | 3-link Reacher | Half-Cheetah | 7-DOF Reacher |
|-----------------------------|----------------|----------------|--------------|---------------|
| Number of training data $N$ | 100            | 250            | 10,000       | 2,000         |
| Number of test data $M_1$   | 500,000        | 1,000,000      | 2,000,000    | 2,000,000     |

**Table 2**

2-link Reacher: RMSEs for one-step prediction (mean $\pm$ 95% CI). Bold values indicate the best performance.

| Models           | GP-PI-K                  | GP-PI-M                  | GP                | GP-NN-M           | GP-NN-K           | PM                 | AutoIP            | DE                |
|------------------|--------------------------|--------------------------|-------------------|-------------------|-------------------|--------------------|-------------------|-------------------|
| $\cos \theta_1$  | 0.011 $\pm$ 0.002        | <b>0.009</b> $\pm$ 0.001 | 0.060 $\pm$ 0.014 | 0.132 $\pm$ 0.028 | 0.104 $\pm$ 0.030 | 0.030 $\pm$ 3.9e-5 | 0.016 $\pm$ 0.002 | 0.351 $\pm$ 0.039 |
| $\cos \theta_2$  | 0.006 $\pm$ 0.001        | <b>0.005</b> $\pm$ 0.001 | 0.052 $\pm$ 0.017 | 0.105 $\pm$ 0.029 | 0.108 $\pm$ 0.033 | 0.025 $\pm$ 1.1e-4 | 0.010 $\pm$ 0.001 | 0.349 $\pm$ 0.049 |
| $\sin \theta_1$  | 0.008 $\pm$ 0.003        | <b>0.006</b> $\pm$ 0.001 | 0.062 $\pm$ 0.016 | 0.087 $\pm$ 0.019 | 0.077 $\pm$ 0.022 | 0.040 $\pm$ 4.0e-5 | 0.010 $\pm$ 0.003 | 0.391 $\pm$ 0.045 |
| $\sin \theta_2$  | <b>0.003</b> $\pm$ 0.001 | 0.004 $\pm$ 0.001        | 0.046 $\pm$ 0.007 | 0.092 $\pm$ 0.013 | 0.079 $\pm$ 0.016 | 0.044 $\pm$ 9.4e-5 | 0.006 $\pm$ 0.004 | 0.351 $\pm$ 0.033 |
| $\dot{\theta}_1$ | <b>0.013</b> $\pm$ 0.003 | 0.014 $\pm$ 0.001        | 0.059 $\pm$ 0.006 | 0.112 $\pm$ 0.019 | 0.101 $\pm$ 0.016 | 0.693 $\pm$ 0.003  | 0.017 $\pm$ 0.003 | 1.081 $\pm$ 0.093 |
| $\dot{\theta}_2$ | 0.012 $\pm$ 0.002        | <b>0.010</b> $\pm$ 0.001 | 0.052 $\pm$ 0.007 | 0.132 $\pm$ 0.017 | 0.119 $\pm$ 0.014 | 0.309 $\pm$ 0.007  | 0.016 $\pm$ 0.003 | 1.023 $\pm$ 0.099 |

**Table 3**

3-link Reacher: RMSEs for one-step prediction.

| Models           | GP-PI-K                  | GP-PI-M                  | GP                | GP-NN-M           | GP-NN-K           | PM                 | AutoIP            | DE                |
|------------------|--------------------------|--------------------------|-------------------|-------------------|-------------------|--------------------|-------------------|-------------------|
| $\cos \theta_1$  | 0.010 $\pm$ 0.001        | <b>0.009</b> $\pm$ 0.001 | 0.028 $\pm$ 0.008 | 0.042 $\pm$ 0.006 | 0.038 $\pm$ 0.007 | 0.012 $\pm$ 1.1e-4 | 0.013 $\pm$ 0.001 | 0.079 $\pm$ 0.018 |
| $\cos \theta_2$  | 0.009 $\pm$ 0.003        | <b>0.007</b> $\pm$ 0.002 | 0.029 $\pm$ 0.006 | 0.051 $\pm$ 0.010 | 0.047 $\pm$ 0.011 | 0.013 $\pm$ 1.2e-4 | 0.015 $\pm$ 0.002 | 0.084 $\pm$ 0.020 |
| $\cos \theta_3$  | 0.011 $\pm$ 0.003        | <b>0.008</b> $\pm$ 0.001 | 0.025 $\pm$ 0.006 | 0.035 $\pm$ 0.007 | 0.032 $\pm$ 0.008 | 0.016 $\pm$ 1.3e-5 | 0.014 $\pm$ 0.003 | 0.078 $\pm$ 0.013 |
| $\sin \theta_1$  | 0.012 $\pm$ 0.001        | <b>0.008</b> $\pm$ 0.001 | 0.023 $\pm$ 0.007 | 0.028 $\pm$ 0.007 | 0.024 $\pm$ 0.006 | 0.022 $\pm$ 1.5e-4 | 0.016 $\pm$ 0.003 | 0.120 $\pm$ 0.019 |
| $\sin \theta_2$  | 0.012 $\pm$ 0.003        | <b>0.008</b> $\pm$ 0.002 | 0.032 $\pm$ 0.012 | 0.036 $\pm$ 0.007 | 0.028 $\pm$ 0.009 | 0.023 $\pm$ 1.5e-4 | 0.017 $\pm$ 0.002 | 0.122 $\pm$ 0.020 |
| $\sin \theta_3$  | 0.009 $\pm$ 0.001        | <b>0.007</b> $\pm$ 0.001 | 0.019 $\pm$ 0.006 | 0.026 $\pm$ 0.008 | 0.021 $\pm$ 0.006 | 0.027 $\pm$ 2.2e-5 | 0.015 $\pm$ 0.003 | 0.108 $\pm$ 0.017 |
| $\dot{\theta}_1$ | <b>0.014</b> $\pm$ 0.003 | 0.027 $\pm$ 0.007        | 0.105 $\pm$ 0.027 | 0.092 $\pm$ 0.030 | 0.089 $\pm$ 0.027 | 0.296 $\pm$ 0.018  | 0.026 $\pm$ 0.006 | 0.157 $\pm$ 0.021 |
| $\dot{\theta}_2$ | <b>0.020</b> $\pm$ 0.004 | 0.043 $\pm$ 0.009        | 0.107 $\pm$ 0.031 | 0.094 $\pm$ 0.026 | 0.092 $\pm$ 0.030 | 0.284 $\pm$ 0.019  | 0.060 $\pm$ 0.007 | 0.159 $\pm$ 0.021 |
| $\dot{\theta}_3$ | <b>0.037</b> $\pm$ 0.005 | 0.046 $\pm$ 0.008        | 0.071 $\pm$ 0.016 | 0.069 $\pm$ 0.014 | 0.067 $\pm$ 0.019 | 0.351 $\pm$ 0.003  | 0.053 $\pm$ 0.006 | 0.192 $\pm$ 0.034 |

- (1) Incorporating prior knowledge via either the kernel function (GP-PI-K) or the mean function (GP-PI-M) significantly improves the extrapolation capability of GPs. This explains why both GP-PI-K and GP-PI-M consistently outperform GP, GP-NN-M, and GP-NN-K, which do not leverage prior knowledge.
- (2) For the 2-link Reacher and 3-link Reacher systems, AutoIP benefits from the use of ODE-based physical prior knowledge and therefore performs better than the standard zero-mean GP model (GP), which is the strongest non-prior GP baseline in these systems. AutoIP is adapted by directly constructing continuous-time training targets from the ground-truth system and mapping the learned continuous-time vector field to next-state prediction using RK4, thereby avoiding first-order finite-difference derivative approximation. Nevertheless, AutoIP remains worse than GP-PI-K and GP-PI-M. A plausible reason is that AutoIP incorporates physical knowledge through derivative constraints in a continuous-time formulation, whereas GP-PI-K and GP-PI-M directly exploit the discretized prior model for the discrete-time next-state prediction task.
- (3) For relatively simple systems such as the 2-link Reacher and 3-link Reacher, introducing NNs into the mean or kernel function leads to unnecessary complexity. As a result, GP-NN-M and GP-NN-K perform worse than GP.
- (4) In contrast, for more complex systems like the Half-Cheetah and 7-DOF Reacher, the added representational capacity of NNs enables GP-NN-M and GP-NN-K to better capture complex dynamics, resulting in improved performance over GP.
- (5) DE performs substantially worse than the GP-based methods, especially in the small-data regime. This is consistent with the general observation that GP models are often more sample-efficient and more robust than purely NN-based models when training data are limited.
- (6) PM exhibits the largest prediction errors due to discrepancies between the true dynamics and the prior models, such as unmodeled friction, damping, or geometric mismatches.

### 5.1.2. Uncertainty estimation and calibration

Tables 6–9 summarize the MNLPs across all systems, where lower values indicate better uncertainty estimation. The observed performance rankings are summarized as follows:

- 2-link Reacher and 3-link Reacher (simple systems):

$$\text{GP-PI-K} > \text{AutoIP} > \text{GP} > \text{GP-PI-M} > \text{GP-NN-K} > \text{GP-NN-M} > \text{DE}, \quad (59)$$

- Half-Cheetah and 7-DOF Reacher (complex systems):

$$\text{GP-PI-K} > \text{GP-NN-K} > \text{GP} > \text{GP-PI-M} > \text{GP-NN-M} > \text{DE}. \quad (60)$$

The MNLP results can be interpreted as follows:

- (1) GP-PI-M and GP-NN-M incorporate explicit mean functions (i.e., prior models or NNs), whereas GP-PI-K, GP, and GP-NN-K adopt zero-mean formulations. In our experiments, the models with explicit mean functions tend to yield more overconfident uncertainty estimates, as also reflected by the empirical coverage and reliability-diagram results presented later. A plausible explanation is that introducing an explicit mean function changes the residual structure to be modeled by the GP, which in turn affects hyperparameter learning and may lead to poorer uncertainty calibration.
- (2) GP-PI-K consistently achieves the lowest MNLPs across all systems. This indicates that GP-PI-K provides the best overall balance between prediction accuracy and uncertainty quality. Since MNLP depends jointly on the predictive mean and predictive variance (as shown in Eq. (54)), the strong MNLP performance of GP-PI-K suggests that its uncertainty estimates are not only sharp but also better aligned with the observed prediction errors.
- (3) For the 2-link Reacher and 3-link Reacher systems, AutoIP outperforms GP, GP-PI-M, GP-NN-K, GP-NN-M, and DE in terms of MNLP. This suggests that incorporating ODE-based prior knowledge improves uncertainty estimation in these simple systems. However,

**Table 4**  
Half-Cheetah: RMSEs for one-step prediction.

| Models                   | GP-PI-K              | GP-PI-M              | GP          | GP-NN-M     | GP-NN-K     | PM           | DE          |
|--------------------------|----------------------|----------------------|-------------|-------------|-------------|--------------|-------------|
| $\frac{x_{t+1}-x_t}{dt}$ | <b>0.206</b> ±0.012  | 0.231±0.009          | 0.467±0.017 | 0.393±0.014 | 0.403±0.016 | 0.611±0.061  | 2.567±0.017 |
| $z$                      | <b>0.007</b> ±2.9e-4 | 0.008±2.9e-4         | 0.017±0.001 | 0.014±0.001 | 0.012±0.002 | 0.020±0.001  | 0.023±0.002 |
| $\cos \theta_1$          | 0.013±0.001          | <b>0.012</b> ±0.001  | 0.016±0.002 | 0.014±0.002 | 0.013±0.001 | 0.048±0.006  | 0.050±0.004 |
| $\cos \theta_2$          | <b>0.008</b> ±0.001  | 0.010±2.9e-4         | 0.027±0.003 | 0.019±0.001 | 0.017±0.001 | 0.022±0.001  | 0.030±0.003 |
| $\cos \theta_3$          | <b>0.008</b> ±0.001  | 0.009±0.001          | 0.019±0.002 | 0.023±0.001 | 0.020±0.002 | 0.022±0.001  | 0.020±0.001 |
| $\cos \theta_4$          | <b>0.006</b> ±2.9e-4 | 0.007±2.9e-4         | 0.015±0.002 | 0.015±0.001 | 0.014±0.002 | 0.016±0.001  | 0.011±0.002 |
| $\cos \theta_5$          | <b>0.013</b> ±0.001  | <b>0.013</b> ±0.001  | 0.025±0.003 | 0.025±0.002 | 0.023±0.002 | 0.032±0.002  | 0.027±0.002 |
| $\cos \theta_6$          | <b>0.012</b> ±0.001  | 0.013±0.001          | 0.025±0.002 | 0.019±0.001 | 0.019±0.001 | 0.030±0.001  | 0.026±0.003 |
| $\cos \theta_7$          | <b>0.007</b> ±3.0e-4 | <b>0.007</b> ±4.2e-4 | 0.015±0.001 | 0.015±0.001 | 0.014±0.001 | 0.018±4.9e-4 | 0.033±0.001 |
| $\sin \theta_1$          | <b>0.014</b> ±0.001  | 0.015±0.001          | 0.026±0.002 | 0.022±0.001 | 0.018±0.002 | 0.035±0.001  | 0.028±0.002 |
| $\sin \theta_2$          | <b>0.017</b> ±0.001  | 0.019±0.002          | 0.029±0.003 | 0.030±0.002 | 0.030±0.001 | 0.054±0.002  | 0.050±0.003 |
| $\sin \theta_3$          | <b>0.018</b> ±0.002  | 0.020±0.002          | 0.043±0.003 | 0.041±0.003 | 0.037±0.002 | 0.051±0.002  | 0.061±0.003 |
| $\sin \theta_4$          | 0.019±0.002          | <b>0.018</b> ±0.002  | 0.037±0.003 | 0.027±0.003 | 0.023±0.001 | 0.045±0.001  | 0.043±0.002 |
| $\sin \theta_5$          | <b>0.028</b> ±0.002  | 0.031±0.001          | 0.041±0.003 | 0.043±0.002 | 0.041±0.004 | 0.073±0.005  | 0.053±0.002 |
| $\sin \theta_6$          | <b>0.027</b> ±0.001  | 0.029±0.001          | 0.053±0.003 | 0.042±0.002 | 0.041±0.003 | 0.077±0.004  | 0.058±0.003 |
| $\sin \theta_7$          | <b>0.024</b> ±0.001  | 0.025±0.001          | 0.043±0.002 | 0.040±0.002 | 0.038±0.002 | 0.057±0.003  | 0.063±0.004 |
| $\dot{x}$                | <b>0.219</b> ±0.013  | 0.244±0.010          | 0.640±0.025 | 0.526±0.035 | 0.450±0.027 | 0.694±0.048  | 1.312±0.016 |
| $\dot{z}$                | <b>0.182</b> ±0.006  | 0.189±0.004          | 0.439±0.006 | 0.390±0.011 | 0.383±0.010 | 0.521±0.012  | 0.870±0.004 |
| $\dot{\theta}_1$         | <b>0.387</b> ±0.011  | 0.441±0.006          | 0.779±0.039 | 0.624±0.035 | 0.603±0.036 | 0.901±0.018  | 0.922±0.021 |
| $\dot{\theta}_2$         | <b>0.735</b> ±0.012  | 0.788±0.008          | 1.713±0.046 | 1.406±0.043 | 1.197±0.046 | 1.768±0.031  | 1.949±0.020 |
| $\dot{\theta}_3$         | <b>1.027</b> ±0.008  | 1.183±0.006          | 2.071±0.017 | 1.807±0.023 | 1.542±0.016 | 1.869±0.027  | 2.866±0.073 |
| $\dot{\theta}_4$         | <b>1.002</b> ±0.005  | 1.004±0.004          | 1.891±0.021 | 1.626±0.028 | 1.491±0.024 | 1.997±0.038  | 2.294±0.044 |
| $\dot{\theta}_5$         | <b>1.003</b> ±0.027  | 1.045±0.024          | 2.426±0.126 | 1.902±0.106 | 1.816±0.130 | 3.252±0.070  | 3.535±0.060 |
| $\dot{\theta}_6$         | 1.397±0.040          | <b>1.380</b> ±0.035  | 2.463±0.092 | 2.212±0.078 | 2.028±0.088 | 2.961±0.047  | 3.002±0.043 |
| $\dot{\theta}_7$         | <b>1.065</b> ±0.010  | 1.086±0.008          | 1.745±0.053 | 1.627±0.067 | 1.661±0.061 | 2.121±0.032  | 2.259±0.057 |

**Table 5**  
7-DOF Reacher: RMSEs for one-step prediction.

| Models           | GP-PI-K              | GP-PI-M              | GP                   | GP-NN-M              | GP-NN-K              | PM           | DE                   |
|------------------|----------------------|----------------------|----------------------|----------------------|----------------------|--------------|----------------------|
| $\cos \theta_1$  | <b>0.001</b> ±2.3e-5 | <b>0.001</b> ±9.4e-6 | <b>0.001</b> ±9.3e-5 | <b>0.001</b> ±3.8e-5 | <b>0.001</b> ±1.2e-4 | 0.003±4.6e-6 | <b>0.001</b> ±2.3e-5 |
| $\cos \theta_2$  | 0.006±0.001          | <b>0.005</b> ±0.001  | 0.010±0.002          | 0.008±0.001          | 0.007±0.001          | 0.034±2.4e-4 | 0.011±0.002          |
| $\cos \theta_3$  | <b>0.011</b> ±0.002  | 0.012±0.001          | 0.028±0.004          | 0.023±0.003          | 0.018±0.004          | 0.054±0.002  | 0.055±0.004          |
| $\cos \theta_4$  | <b>0.014</b> ±0.002  | 0.015±0.001          | 0.031±0.005          | 0.027±0.006          | 0.029±0.007          | 0.069±0.002  | 0.363±0.015          |
| $\cos \theta_5$  | 0.008±0.001          | <b>0.007</b> ±0.002  | 0.012±0.003          | 0.011±0.003          | 0.008±0.002          | 0.048±0.002  | 0.127±0.005          |
| $\cos \theta_6$  | <b>0.005</b> ±0.001  | 0.006±0.002          | 0.009±0.002          | 0.011±0.002          | 0.009±0.002          | 0.028±0.001  | 0.137±0.004          |
| $\cos \theta_7$  | <b>0.008</b> ±0.002  | 0.009±0.002          | 0.016±0.003          | 0.013±0.003          | 0.011±0.003          | 0.126±0.002  | 0.105±0.005          |
| $\sin \theta_1$  | <b>0.005</b> ±2.3e-4 | 0.007±2.5e-4         | 0.009±0.002          | 0.010±0.002          | 0.009±0.002          | 0.007±3.0e-4 | 0.018±0.004          |
| $\sin \theta_2$  | 0.017±0.003          | <b>0.014</b> ±0.002  | 0.031±0.005          | 0.028±0.004          | 0.024±0.005          | 0.018±0.001  | 0.018±0.005          |
| $\sin \theta_3$  | <b>0.006</b> ±0.001  | 0.007±0.001          | 0.010±0.003          | 0.008±0.003          | 0.009±0.003          | 0.060±0.002  | 0.055±0.004          |
| $\sin \theta_4$  | <b>0.005</b> ±4.7e-4 | 0.006±2.8e-4         | 0.009±0.002          | 0.007±0.001          | 0.006±0.001          | 0.037±0.001  | 0.036±0.004          |
| $\sin \theta_5$  | <b>0.011</b> ±0.003  | 0.013±0.005          | 0.026±0.006          | 0.022±0.005          | 0.020±0.006          | 0.064±0.003  | 0.140±0.005          |
| $\sin \theta_6$  | <b>0.004</b> ±4.7e-4 | 0.006±0.001          | 0.010±0.001          | 0.008±0.002          | 0.009±0.001          | 0.040±0.001  | 0.088±0.005          |
| $\sin \theta_7$  | <b>0.010</b> ±0.003  | 0.011±0.003          | 0.019±0.004          | 0.015±0.003          | 0.012±0.003          | 0.074±0.006  | 0.100±0.005          |
| $\dot{\theta}_1$ | <b>0.009</b> ±0.003  | 0.011±0.002          | 0.012±0.002          | 0.010±0.003          | 0.010±0.003          | 0.033±4.2e-4 | 0.021±0.003          |
| $\dot{\theta}_2$ | 0.011±0.002          | <b>0.008</b> ±0.001  | 0.015±0.001          | 0.013±0.003          | 0.014±0.002          | 0.047±4.2e-4 | 0.026±0.004          |
| $\dot{\theta}_3$ | 0.066±0.004          | <b>0.062</b> ±0.004  | 0.083±0.009          | 0.077±0.008          | 0.079±0.007          | 0.156±0.006  | 0.141±0.008          |
| $\dot{\theta}_4$ | 0.053±0.006          | <b>0.049</b> ±0.005  | 0.075±0.008          | 0.067±0.007          | 0.073±0.007          | 0.132±0.002  | 0.101±0.006          |
| $\dot{\theta}_5$ | <b>0.031</b> ±0.006  | 0.036±0.005          | 0.056±0.011          | 0.048±0.008          | 0.042±0.009          | 0.145±0.002  | 0.187±0.008          |
| $\dot{\theta}_6$ | <b>0.135</b> ±0.010  | 0.160±0.008          | 0.189±0.017          | 0.181±0.015          | 0.176±0.020          | 0.335±0.013  | 0.423±0.013          |
| $\dot{\theta}_7$ | <b>0.173</b> ±0.015  | 0.184±0.013          | 0.262±0.026          | 0.213±0.021          | 0.200±0.023          | 0.508±0.010  | 0.568±0.020          |

AutoIP still performs worse than GP-PI-K, which is consistent with the one-step prediction results in [Section 5.1.1](#).

- (4) For simple systems such as the 2-link Reacher and 3-link Reacher, GP-PI-M achieves extremely high one-step prediction accuracy. This compensates, to some extent, for its underestimation of uncertainty, enabling it to outperform GP-NN-K in the MNLP metric.
- (5) For simple systems, GP outperforms GP-NN-K; in contrast, for complex systems such as the Half-Cheetah and 7-DOF Reacher, GP-NN-K outperforms GP. This is consistent with the results discussed in [Section 5.1.1](#).
- (6) Due to the incorporation of prior knowledge, GP-PI-M consistently outperforms GP-NN-M.
- (7) DE performs the worst in terms of MNLP. This is consistent with its poor one-step prediction accuracy and further suggests that, in the relatively small-data setting considered here, purely NN-based probabilistic models are less reliable than GP-based models for uncertainty-aware dynamics modeling.

However, MNLP alone does not fully characterize uncertainty calibration. We therefore further evaluate calibration using empirical coverage and reliability diagrams. We report empirical coverages at the 68% and 95% nominal levels in [Table 10](#) and visualize the corresponding reliability diagrams in [Fig. 3](#). Our goal is not only to improve prediction accuracy, but also to preserve, as much as possible, the favorable uncertainty quantification behavior of standard GPs. Since empirical coverages closer to the nominal levels indicate better calibration, larger coverage values should not be interpreted as better uncertainty estimates. The results show that GP generally achieves empirical coverages closest to the nominal levels. GP-PI-K often over-covers relative to the nominal 68% and 95% levels, indicating that its predictive intervals are somewhat conservative. Therefore, we do not claim that GP-PI-K is perfectly calibrated or uniformly better calibrated than GP.

Rather, the main observation is that GP-PI-K substantially improves prediction accuracy while retaining GP-like uncertainty behavior. In particular, compared with the mean-function-based methods, especially

**Table 6**  
2-link Reacher: MNLPs (mean $\pm$ 95% CI).

| Models           | GP-PI-K                   | GP-PI-M            | GP                 | GP-NN-M            | GP-NN-K            | AutoIP             | DE                |
|------------------|---------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|-------------------|
| $\cos \theta_1$  | <b>-3.103</b> $\pm$ 0.033 | -1.678 $\pm$ 0.025 | -2.184 $\pm$ 0.124 | 0.282 $\pm$ 0.184  | -0.425 $\pm$ 0.155 | -2.814 $\pm$ 0.062 | 1.185 $\pm$ 0.429 |
| $\cos \theta_2$  | <b>-3.513</b> $\pm$ 0.004 | -1.553 $\pm$ 0.003 | -2.215 $\pm$ 0.185 | -0.037 $\pm$ 0.210 | -0.662 $\pm$ 0.174 | -2.639 $\pm$ 0.059 | 1.350 $\pm$ 0.494 |
| $\sin \theta_1$  | <b>-3.202</b> $\pm$ 0.036 | -1.467 $\pm$ 0.029 | -2.142 $\pm$ 0.144 | 0.654 $\pm$ 0.193  | -0.516 $\pm$ 0.168 | -2.842 $\pm$ 0.096 | 1.404 $\pm$ 0.416 |
| $\sin \theta_2$  | <b>-3.599</b> $\pm$ 0.005 | -1.259 $\pm$ 0.002 | -2.227 $\pm$ 0.147 | -0.110 $\pm$ 0.322 | -0.703 $\pm$ 0.290 | -2.944 $\pm$ 0.059 | 0.842 $\pm$ 0.221 |
| $\dot{\theta}_1$ | <b>-2.638</b> $\pm$ 0.043 | -1.369 $\pm$ 0.038 | -2.077 $\pm$ 0.084 | 1.096 $\pm$ 0.253  | -0.620 $\pm$ 0.182 | -2.421 $\pm$ 0.083 | 1.549 $\pm$ 0.122 |
| $\dot{\theta}_2$ | <b>-2.744</b> $\pm$ 0.047 | -1.270 $\pm$ 0.039 | -2.100 $\pm$ 0.064 | 2.391 $\pm$ 0.192  | -0.599 $\pm$ 0.135 | -2.387 $\pm$ 0.082 | 1.921 $\pm$ 0.212 |

**Table 7**  
3-link Reacher: MNLPs.

| Models           | GP-PI-K                   | GP-PI-M            | GP                 | GP-NN-M            | GP-NN-K            | AutoIP             | DE                 |
|------------------|---------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| $\cos \theta_1$  | <b>-3.415</b> $\pm$ 0.060 | -2.281 $\pm$ 0.052 | -2.552 $\pm$ 0.083 | -0.762 $\pm$ 0.126 | -2.061 $\pm$ 0.135 | -2.909 $\pm$ 0.096 | 3.310 $\pm$ 0.980  |
| $\cos \theta_2$  | <b>-3.215</b> $\pm$ 0.053 | -2.484 $\pm$ 0.049 | -2.541 $\pm$ 0.074 | -0.882 $\pm$ 0.132 | -2.058 $\pm$ 0.109 | -2.792 $\pm$ 0.121 | 4.393 $\pm$ 1.019  |
| $\cos \theta_3$  | <b>-3.358</b> $\pm$ 0.061 | -2.277 $\pm$ 0.080 | -2.598 $\pm$ 0.064 | -0.842 $\pm$ 0.118 | -2.152 $\pm$ 0.090 | -2.933 $\pm$ 0.119 | 2.589 $\pm$ 0.706  |
| $\sin \theta_1$  | <b>-3.429</b> $\pm$ 0.061 | -2.384 $\pm$ 0.051 | -2.608 $\pm$ 0.050 | -1.020 $\pm$ 0.100 | -2.150 $\pm$ 0.047 | -2.840 $\pm$ 0.116 | 1.785 $\pm$ 0.377  |
| $\sin \theta_2$  | <b>-3.433</b> $\pm$ 0.049 | -2.290 $\pm$ 0.050 | -2.430 $\pm$ 0.126 | -1.100 $\pm$ 0.146 | -2.042 $\pm$ 0.112 | -2.622 $\pm$ 0.139 | 2.319 $\pm$ 0.494  |
| $\sin \theta_3$  | <b>-3.422</b> $\pm$ 0.054 | -2.488 $\pm$ 0.057 | -2.608 $\pm$ 0.057 | -1.132 $\pm$ 0.105 | -2.586 $\pm$ 0.080 | -2.968 $\pm$ 0.134 | 0.650 $\pm$ 0.248  |
| $\dot{\theta}_1$ | <b>-3.068</b> $\pm$ 0.062 | -2.111 $\pm$ 0.084 | -1.962 $\pm$ 0.113 | -1.401 $\pm$ 0.196 | -1.493 $\pm$ 0.176 | -2.729 $\pm$ 0.197 | -0.469 $\pm$ 0.181 |
| $\dot{\theta}_2$ | <b>-2.913</b> $\pm$ 0.091 | -1.685 $\pm$ 0.100 | -1.794 $\pm$ 0.195 | -1.374 $\pm$ 0.194 | -1.579 $\pm$ 0.183 | -2.417 $\pm$ 0.147 | -0.188 $\pm$ 0.267 |
| $\dot{\theta}_3$ | <b>-2.801</b> $\pm$ 0.058 | -1.970 $\pm$ 0.095 | -2.048 $\pm$ 0.076 | -1.454 $\pm$ 0.251 | -1.844 $\pm$ 0.130 | -2.493 $\pm$ 0.128 | -0.142 $\pm$ 0.215 |

**Table 8**  
Half-Cheetah: MNLPs.

| Models                   | GP-PI-K                   | GP-PI-M            | GP                 | GP-NN-M            | GP-NN-K            | DE                 |
|--------------------------|---------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| $\frac{x_{t+1}-x_t}{dt}$ | <b>-0.885</b> $\pm$ 0.024 | 0.592 $\pm$ 0.022  | 0.135 $\pm$ 0.040  | 1.254 $\pm$ 0.093  | 0.116 $\pm$ 0.032  | 11.805 $\pm$ 2.513 |
| $z$                      | <b>-3.676</b> $\pm$ 0.036 | -1.327 $\pm$ 0.025 | -2.673 $\pm$ 0.087 | -0.499 $\pm$ 0.116 | -2.476 $\pm$ 0.096 | 0.422 $\pm$ 0.100  |
| $\cos \theta_1$          | <b>-3.760</b> $\pm$ 0.032 | -1.448 $\pm$ 0.025 | -3.061 $\pm$ 0.044 | -0.874 $\pm$ 0.060 | -3.127 $\pm$ 0.052 | -0.147 $\pm$ 0.111 |
| $\cos \theta_2$          | <b>-3.609</b> $\pm$ 0.006 | -1.325 $\pm$ 0.015 | -1.857 $\pm$ 0.052 | -0.383 $\pm$ 0.082 | -2.022 $\pm$ 0.060 | 0.100 $\pm$ 0.132  |
| $\cos \theta_3$          | <b>-3.425</b> $\pm$ 0.010 | -1.205 $\pm$ 0.014 | -1.978 $\pm$ 0.056 | -0.359 $\pm$ 0.128 | -1.986 $\pm$ 0.053 | 0.262 $\pm$ 0.095  |
| $\cos \theta_4$          | <b>-3.965</b> $\pm$ 0.002 | -1.333 $\pm$ 0.004 | -2.961 $\pm$ 0.027 | -0.126 $\pm$ 0.053 | -3.013 $\pm$ 0.023 | 0.282 $\pm$ 0.099  |
| $\cos \theta_5$          | <b>-3.033</b> $\pm$ 0.024 | -1.536 $\pm$ 0.023 | -2.079 $\pm$ 0.122 | -0.824 $\pm$ 0.135 | -2.091 $\pm$ 0.160 | 0.496 $\pm$ 0.171  |
| $\cos \theta_6$          | <b>-3.146</b> $\pm$ 0.013 | -1.389 $\pm$ 0.020 | -2.186 $\pm$ 0.036 | -0.773 $\pm$ 0.071 | -2.115 $\pm$ 0.022 | -0.141 $\pm$ 0.196 |
| $\cos \theta_7$          | <b>-3.582</b> $\pm$ 0.021 | -1.690 $\pm$ 0.019 | -2.680 $\pm$ 0.025 | -0.844 $\pm$ 0.019 | -2.745 $\pm$ 0.025 | -0.406 $\pm$ 0.071 |
| $\sin \theta_1$          | <b>-3.099</b> $\pm$ 0.022 | -1.676 $\pm$ 0.014 | -2.162 $\pm$ 0.055 | -0.648 $\pm$ 0.075 | -2.490 $\pm$ 0.041 | -0.022 $\pm$ 0.125 |
| $\sin \theta_2$          | <b>-2.757</b> $\pm$ 0.010 | -1.485 $\pm$ 0.020 | -2.031 $\pm$ 0.050 | -0.570 $\pm$ 0.061 | -2.165 $\pm$ 0.041 | 0.316 $\pm$ 0.191  |
| $\sin \theta_3$          | <b>-2.663</b> $\pm$ 0.008 | -1.554 $\pm$ 0.006 | -2.044 $\pm$ 0.051 | -0.825 $\pm$ 0.074 | -1.821 $\pm$ 0.042 | 0.320 $\pm$ 0.193  |
| $\sin \theta_4$          | <b>-2.683</b> $\pm$ 0.011 | -1.366 $\pm$ 0.010 | -1.855 $\pm$ 0.010 | -0.616 $\pm$ 0.032 | -1.815 $\pm$ 0.020 | 0.370 $\pm$ 0.121  |
| $\sin \theta_5$          | <b>-2.287</b> $\pm$ 0.017 | -1.025 $\pm$ 0.025 | -1.457 $\pm$ 0.119 | -0.796 $\pm$ 0.318 | -1.710 $\pm$ 0.142 | 0.636 $\pm$ 0.151  |
| $\sin \theta_6$          | <b>-2.285</b> $\pm$ 0.010 | -0.843 $\pm$ 0.015 | -1.396 $\pm$ 0.069 | -0.545 $\pm$ 0.071 | -1.625 $\pm$ 0.066 | 0.110 $\pm$ 0.117  |
| $\sin \theta_7$          | <b>-2.338</b> $\pm$ 0.010 | -1.221 $\pm$ 0.017 | -1.519 $\pm$ 0.051 | -0.474 $\pm$ 0.067 | -1.769 $\pm$ 0.044 | 0.080 $\pm$ 0.176  |
| $\dot{x}$                | <b>-0.438</b> $\pm$ 0.017 | 0.207 $\pm$ 0.019  | -0.232 $\pm$ 0.212 | 0.976 $\pm$ 0.327  | -0.119 $\pm$ 0.274 | 9.200 $\pm$ 1.971  |
| $\dot{z}$                | <b>-0.376</b> $\pm$ 0.024 | 0.756 $\pm$ 0.023  | 0.414 $\pm$ 0.024  | 1.258 $\pm$ 0.034  | 0.315 $\pm$ 0.029  | 1.538 $\pm$ 0.145  |
| $\dot{\theta}_1$         | <b>0.110</b> $\pm$ 0.010  | 1.264 $\pm$ 0.016  | 0.779 $\pm$ 0.110  | 1.564 $\pm$ 0.179  | 0.832 $\pm$ 0.159  | 3.779 $\pm$ 0.188  |
| $\dot{\theta}_2$         | <b>0.529</b> $\pm$ 0.011  | 1.466 $\pm$ 0.014  | 1.538 $\pm$ 0.091  | 2.085 $\pm$ 0.119  | 1.434 $\pm$ 0.095  | 3.758 $\pm$ 0.418  |
| $\dot{\theta}_3$         | <b>0.684</b> $\pm$ 0.008  | 2.197 $\pm$ 0.002  | 1.715 $\pm$ 0.059  | 2.215 $\pm$ 0.083  | 1.867 $\pm$ 0.050  | 5.188 $\pm$ 0.414  |
| $\dot{\theta}_4$         | <b>0.691</b> $\pm$ 0.011  | 1.992 $\pm$ 0.007  | 1.575 $\pm$ 0.016  | 1.861 $\pm$ 0.038  | 1.704 $\pm$ 0.017  | 4.406 $\pm$ 0.544  |
| $\dot{\theta}_5$         | <b>0.693</b> $\pm$ 0.018  | 2.176 $\pm$ 0.014  | 1.752 $\pm$ 0.143  | 2.588 $\pm$ 0.301  | 1.873 $\pm$ 0.192  | 6.731 $\pm$ 1.249  |
| $\dot{\theta}_6$         | <b>0.831</b> $\pm$ 0.021  | 2.344 $\pm$ 0.029  | 1.892 $\pm$ 0.109  | 3.042 $\pm$ 0.211  | 2.022 $\pm$ 0.126  | 4.050 $\pm$ 0.247  |
| $\dot{\theta}_7$         | <b>0.735</b> $\pm$ 0.006  | 1.900 $\pm$ 0.006  | 1.636 $\pm$ 0.051  | 2.324 $\pm$ 0.111  | 1.411 $\pm$ 0.067  | 3.842 $\pm$ 0.318  |

GP-PI-M and GP-NN-M, GP-PI-K avoids severe under-coverage and therefore avoids strongly overconfident predictive intervals. The reliability curves of GP-PI-K also remain predictably close to the diagonal across the four systems, although GP is usually closer in the strict coverage-error sense. These results suggest that GP-PI-K provides a favorable balance between predictive performance and uncertainty quantification, rather than a uniform improvement in calibration over standard GPs.

In addition to empirical coverage and reliability diagrams, Table 11 reports CRPS, IS68, IS95, and RE to further evaluate probabilistic prediction quality and calibration. The results show that GP-PI-K achieves the lowest CRPS, IS68, and IS95 across all four systems, indicating the best overall probabilistic prediction quality when predictive accuracy, sharpness, and calibration are jointly considered. In terms of RE, GP often obtains the smallest value, which is consistent with the empirical coverage results. Nevertheless, GP-PI-K still achieves competitive

RE values compared with GP while substantially improving CRPS and interval scores.

To further examine uncertainty behavior, we visualize the log predictive stds for the low-dimensional systems (2-link Reacher and 3-link Reacher) in Fig. 4. These visualizations depict the uncertainty landscape over the  $[\theta_1, \theta_2]^T$  state space, using 500,000 and 1,000,000 test data points, respectively. As expected for probabilistic models with reliable uncertainty quantification, GP-PI-K, GP, GP-NN-K, and AutoIP exhibit relatively low uncertainty in regions covered by training data and higher uncertainty in unobserved areas. In contrast, GP-PI-M and GP-NN-M show uniformly low uncertainty over much of the state space, making them less able to distinguish between known and unknown regions. This behavior is consistent with the empirical coverage and reliability-diagram results, and further suggests that GP-PI-K preserves the uncertainty behavior of standard GPs more effectively than the mean-function-based baselines.

**Table 9**  
7-DOF Reacher: MNLPs.

| Models           | GP-PI-K                   | GP-PI-M            | GP                 | GP-NN-M           | GP-NN-K            | DE                 |
|------------------|---------------------------|--------------------|--------------------|-------------------|--------------------|--------------------|
| $\cos \theta_1$  | <b>-6.484</b> $\pm 0.110$ | -2.227 $\pm 0.189$ | -4.122 $\pm 0.224$ | 2.845 $\pm 0.659$ | -4.469 $\pm 0.198$ | 3.745 $\pm 0.722$  |
| $\cos \theta_2$  | <b>-6.335</b> $\pm 0.101$ | -2.013 $\pm 0.152$ | -4.221 $\pm 0.170$ | 2.665 $\pm 0.620$ | -4.925 $\pm 0.191$ | 2.514 $\pm 0.669$  |
| $\cos \theta_3$  | <b>-3.807</b> $\pm 0.071$ | -0.292 $\pm 0.119$ | -2.941 $\pm 0.103$ | 2.375 $\pm 0.570$ | -3.289 $\pm 0.145$ | 2.447 $\pm 0.319$  |
| $\cos \theta_4$  | <b>-3.358</b> $\pm 0.058$ | -0.490 $\pm 0.070$ | -2.481 $\pm 0.126$ | 4.760 $\pm 0.708$ | -2.382 $\pm 0.116$ | 28.853 $\pm 4.614$ |
| $\cos \theta_5$  | <b>-3.392</b> $\pm 0.051$ | -0.391 $\pm 0.075$ | -2.685 $\pm 0.092$ | 5.403 $\pm 0.582$ | -2.916 $\pm 0.119$ | 6.336 $\pm 0.845$  |
| $\cos \theta_6$  | <b>-3.525</b> $\pm 0.060$ | -0.533 $\pm 0.088$ | -3.063 $\pm 0.163$ | 2.541 $\pm 0.675$ | -2.822 $\pm 0.154$ | 3.705 $\pm 0.571$  |
| $\cos \theta_7$  | <b>-3.371</b> $\pm 0.080$ | -0.744 $\pm 0.078$ | -2.711 $\pm 0.109$ | 5.375 $\pm 0.965$ | -3.035 $\pm 0.143$ | 6.624 $\pm 0.908$  |
| $\sin \theta_1$  | <b>-5.024</b> $\pm 0.062$ | -1.646 $\pm 0.080$ | -3.485 $\pm 0.070$ | 4.284 $\pm 0.853$ | -3.759 $\pm 0.080$ | 6.692 $\pm 1.182$  |
| $\sin \theta_2$  | <b>-4.816</b> $\pm 0.078$ | -1.254 $\pm 0.086$ | -3.374 $\pm 0.134$ | 4.726 $\pm 1.034$ | -3.648 $\pm 0.103$ | 3.496 $\pm 0.726$  |
| $\sin \theta_3$  | <b>-3.940</b> $\pm 0.066$ | 1.128 $\pm 0.104$  | -3.085 $\pm 0.144$ | 5.513 $\pm 1.111$ | -3.342 $\pm 0.164$ | 6.434 $\pm 1.266$  |
| $\sin \theta_4$  | <b>-4.219</b> $\pm 0.074$ | -0.604 $\pm 0.079$ | -3.546 $\pm 0.131$ | 3.829 $\pm 0.908$ | -3.612 $\pm 0.143$ | 2.457 $\pm 0.800$  |
| $\sin \theta_5$  | <b>-3.560</b> $\pm 0.057$ | -0.349 $\pm 0.068$ | -2.603 $\pm 0.092$ | 4.104 $\pm 0.808$ | -2.993 $\pm 0.078$ | 4.230 $\pm 0.779$  |
| $\sin \theta_6$  | <b>-3.946</b> $\pm 0.048$ | -0.646 $\pm 0.061$ | -3.094 $\pm 0.095$ | 3.222 $\pm 0.643$ | -3.371 $\pm 0.120$ | 3.432 $\pm 0.574$  |
| $\sin \theta_7$  | <b>-3.410</b> $\pm 0.062$ | 0.447 $\pm 0.074$  | -2.856 $\pm 0.121$ | 2.776 $\pm 0.693$ | -2.788 $\pm 0.112$ | 3.894 $\pm 0.604$  |
| $\dot{\theta}_1$ | <b>-3.342</b> $\pm 0.021$ | -0.160 $\pm 0.048$ | -2.736 $\pm 0.028$ | 2.649 $\pm 0.573$ | -3.092 $\pm 0.035$ | 2.675 $\pm 0.805$  |
| $\dot{\theta}_2$ | <b>-3.207</b> $\pm 0.018$ | -0.580 $\pm 0.056$ | -2.658 $\pm 0.032$ | 2.380 $\pm 0.476$ | -2.871 $\pm 0.037$ | 1.938 $\pm 0.585$  |
| $\dot{\theta}_3$ | <b>-1.256</b> $\pm 0.017$ | 0.814 $\pm 0.044$  | -0.927 $\pm 0.043$ | 3.464 $\pm 0.745$ | -1.063 $\pm 0.040$ | 3.523 $\pm 0.748$  |
| $\dot{\theta}_4$ | <b>-1.327</b> $\pm 0.021$ | 0.930 $\pm 0.052$  | -0.950 $\pm 0.045$ | 4.071 $\pm 0.969$ | -1.059 $\pm 0.049$ | 3.899 $\pm 0.672$  |
| $\dot{\theta}_5$ | <b>-2.069</b> $\pm 0.027$ | 1.224 $\pm 0.050$  | -1.896 $\pm 0.039$ | 5.202 $\pm 1.188$ | -1.903 $\pm 0.044$ | 5.462 $\pm 0.646$  |
| $\dot{\theta}_6$ | <b>-0.651</b> $\pm 0.031$ | 1.376 $\pm 0.072$  | -0.529 $\pm 0.058$ | 4.978 $\pm 1.043$ | -0.577 $\pm 0.049$ | 5.930 $\pm 0.666$  |
| $\dot{\theta}_7$ | <b>-1.494</b> $\pm 0.029$ | 1.534 $\pm 0.105$  | -0.974 $\pm 0.042$ | 6.277 $\pm 1.261$ | -1.106 $\pm 0.045$ | 5.576 $\pm 0.868$  |

**Table 10**

Empirical coverages at the 68% (cov68) and 95% (cov95) nominal levels (mean  $\pm$  95% CI). Values closer to the nominal levels indicate better uncertainty calibration.

| (a) 2-link Reacher |                   |                          |                   |                   |                   |                   |                   |
|--------------------|-------------------|--------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                    | GP-PI-K           | GP                       | GP-NN-K           | AutoIP            | GP-PI-M           | GP-NN-M           | DE                |
| cov68              | 0.797 $\pm 0.019$ | <b>0.769</b> $\pm 0.013$ | 0.813 $\pm 0.026$ | 0.858 $\pm 0.021$ | 0.511 $\pm 0.009$ | 0.218 $\pm 0.023$ | 0.476 $\pm 0.029$ |
| cov95              | 0.979 $\pm 0.011$ | <b>0.970</b> $\pm 0.007$ | 0.985 $\pm 0.027$ | 0.986 $\pm 0.008$ | 0.776 $\pm 0.006$ | 0.345 $\pm 0.030$ | 0.682 $\pm 0.027$ |
| (b) 3-link Reacher |                   |                          |                   |                   |                   |                   |                   |
|                    | GP-PI-K           | GP                       | GP-NN-K           | AutoIP            | GP-PI-M           | GP-NN-M           | DE                |
| cov68              | 0.784 $\pm 0.012$ | <b>0.743</b> $\pm 0.019$ | 0.828 $\pm 0.025$ | 0.854 $\pm 0.027$ | 0.524 $\pm 0.010$ | 0.250 $\pm 0.020$ | 0.450 $\pm 0.017$ |
| cov95              | 0.972 $\pm 0.007$ | <b>0.962</b> $\pm 0.013$ | 0.988 $\pm 0.022$ | 0.989 $\pm 0.023$ | 0.715 $\pm 0.007$ | 0.542 $\pm 0.018$ | 0.625 $\pm 0.019$ |
| (c) Half-Cheetah   |                   |                          |                   |                   |                   |                   |                   |
|                    | GP-PI-K           | GP                       | GP-NN-K           | GP-PI-M           | GP-NN-M           | DE                |                   |
| cov68              | 0.723 $\pm 0.024$ | <b>0.698</b> $\pm 0.029$ | 0.773 $\pm 0.028$ | 0.547 $\pm 0.018$ | 0.301 $\pm 0.031$ | 0.428 $\pm 0.015$ |                   |
| cov95              | 0.967 $\pm 0.020$ | <b>0.958</b> $\pm 0.024$ | 0.975 $\pm 0.026$ | 0.762 $\pm 0.015$ | 0.504 $\pm 0.028$ | 0.602 $\pm 0.012$ |                   |
| (d) 7-DOF Reacher  |                   |                          |                   |                   |                   |                   |                   |
|                    | GP-PI-K           | GP                       | GP-NN-K           | GP-PI-M           | GP-NN-M           | DE                |                   |
| cov68              | 0.741 $\pm 0.017$ | <b>0.717</b> $\pm 0.019$ | 0.792 $\pm 0.020$ | 0.450 $\pm 0.015$ | 0.290 $\pm 0.024$ | 0.360 $\pm 0.021$ |                   |
| cov95              | 0.965 $\pm 0.011$ | <b>0.959</b> $\pm 0.014$ | 0.976 $\pm 0.012$ | 0.710 $\pm 0.010$ | 0.535 $\pm 0.020$ | 0.593 $\pm 0.015$ |                   |

### 5.1.3. Multi-step prediction accuracy

Fig. 5 presents the normalized RMSEs for  $L$ -step prediction, where  $L$  ranges from 5 to 20. The curves and error bars represent the means and 95% CIs over 20 random seeds. We summarize the performance rankings as follows:

- 2-link Reacher and 3-link Reacher (simple systems):

$$\text{GP-PI-K} > \text{GP-PI-M} > \text{AutoIP} > \text{GP} > \text{GP-NN-K} > \text{GP-NN-M} > \text{DE, PM}, \quad (61)$$

- Half-Cheetah and 7-DOF Reacher (complex systems):

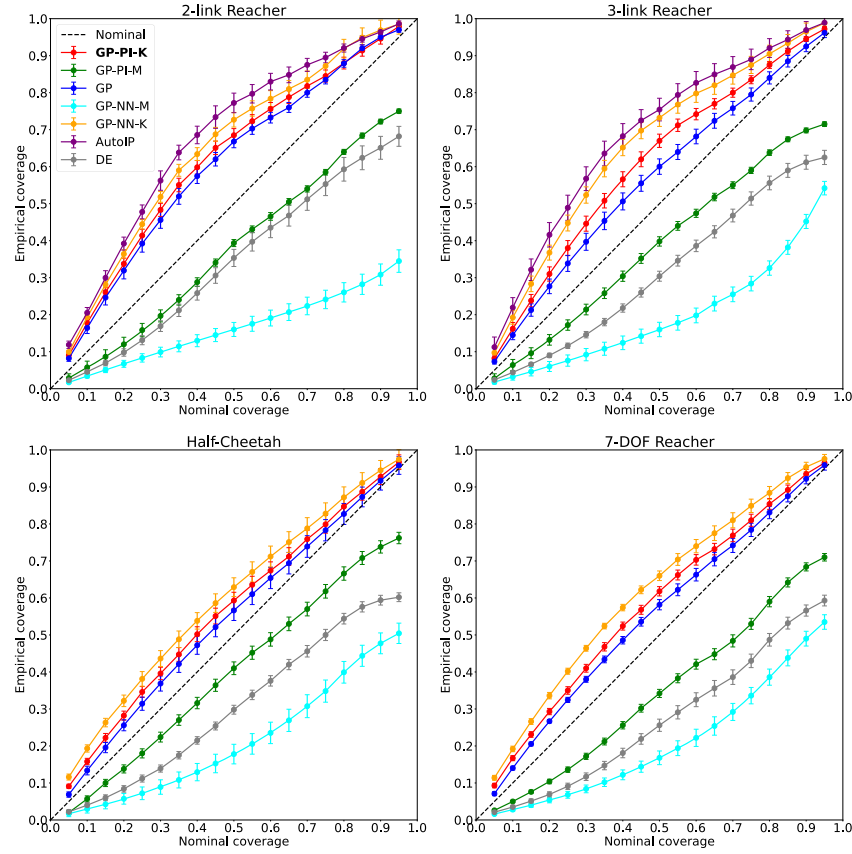
$$\text{GP-PI-K} > \text{GP-NN-K} > \text{GP} > \text{GP-PI-M} > \text{GP-NN-M} > \text{DE} > \text{PM}. \quad (62)$$

The results of multi-step prediction accuracy can be interpreted as follows:

- (1) GP-PI-K consistently achieves the best multi-step prediction accuracy across all four systems. This indicates that incorporating prior knowledge through the kernel function is particularly effective for long-horizon prediction.

- (2) For the 2-link Reacher and 3-link Reacher systems, GP-PI-M also performs strongly, although it is consistently worse than GP-PI-K. AutoIP outperforms GP in these simple systems, but remains inferior to GP-PI-K and GP-PI-M, which is consistent with the one-step prediction results.
- (3) For the Half-Cheetah and 7-DOF Reacher systems, GP-NN-K becomes the second-best method, outperforming both GP and GP-PI-M. This suggests that, for complex systems, NN-based feature learning is more helpful for long-horizon prediction when prior knowledge is not sufficiently accurate to support a mean-function-based formulation.
- (4) In contrast, the multi-step errors of GP-PI-M and GP-NN-M increase more rapidly on the complex systems. This suggests that mean-function-based formulations are more sensitive to model mismatch in recursive prediction. Once the predictive mean is repeatedly fed back into future steps, the effect of an imperfect mean function can accumulate and lead to faster performance degradation.
- (5) DE and PM show relatively poor multi-step performance. In particular, PM deteriorates most severely, indicating that directly using



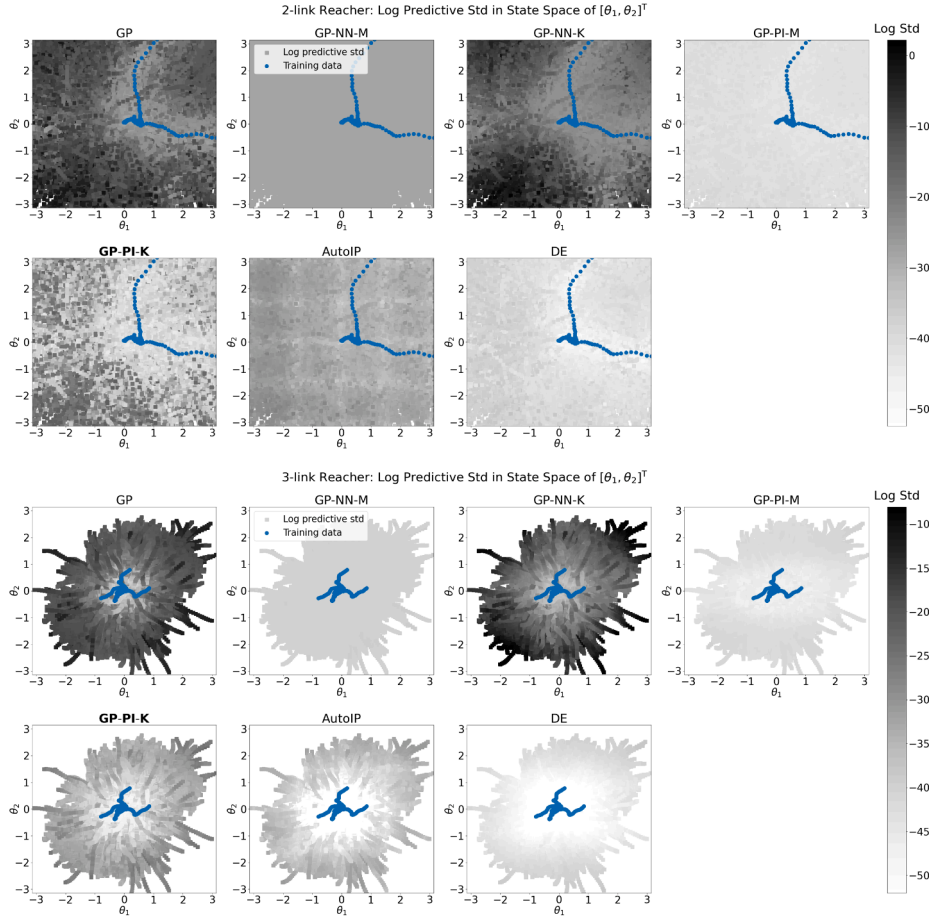


**Fig. 3.** Reliability diagrams. The dashed diagonal indicates ideal calibration. Curves above the diagonal indicate conservative predictive intervals, whereas curves below the diagonal indicate under-coverage. Error bars show 95% CIs over 20 random seeds.

**Table 11**

CRPS, interval scores, and reliability error for probabilistic prediction quality and calibration (mean $\pm$ 95% CI). IS68 and IS95 denote the interval scores at the 68% and 95% nominal levels, respectively. RE denotes reliability error. Lower values are better for all metrics.

| (a) 2-link Reacher |                          |                          |                   |                   |                   |                   |                   |
|--------------------|--------------------------|--------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
|                    | GP-PI-K                  | GP                       | GP-NN-K           | AutoIP            | GP-PI-M           | GP-NN-M           | DE                |
| CRPS               | <b>0.006</b> $\pm$ 0.001 | 0.039 $\pm$ 0.005        | 0.105 $\pm$ 0.029 | 0.016 $\pm$ 0.004 | 0.040 $\pm$ 0.003 | 0.144 $\pm$ 0.020 | 0.373 $\pm$ 0.043 |
| IS68               | <b>0.046</b> $\pm$ 0.004 | 0.222 $\pm$ 0.042        | 0.430 $\pm$ 0.179 | 0.089 $\pm$ 0.011 | 0.251 $\pm$ 0.010 | 0.886 $\pm$ 0.101 | 2.079 $\pm$ 0.251 |
| IS95               | <b>0.088</b> $\pm$ 0.008 | 0.377 $\pm$ 0.086        | 2.543 $\pm$ 0.496 | 0.174 $\pm$ 0.017 | 0.422 $\pm$ 0.021 | 3.913 $\pm$ 0.661 | 4.668 $\pm$ 0.911 |
| RE                 | 0.123 $\pm$ 0.004        | <b>0.108</b> $\pm$ 0.003 | 0.154 $\pm$ 0.004 | 0.180 $\pm$ 0.004 | 0.123 $\pm$ 0.002 | 0.335 $\pm$ 0.003 | 0.160 $\pm$ 0.005 |
| (b) 3-link Reacher |                          |                          |                   |                   |                   |                   |                   |
|                    | GP-PI-K                  | GP                       | GP-NN-K           | AutoIP            | GP-PI-M           | GP-NN-M           | DE                |
| CRPS               | <b>0.011</b> $\pm$ 0.001 | 0.047 $\pm$ 0.010        | 0.063 $\pm$ 0.022 | 0.027 $\pm$ 0.005 | 0.043 $\pm$ 0.002 | 0.088 $\pm$ 0.010 | 0.153 $\pm$ 0.025 |
| IS68               | <b>0.082</b> $\pm$ 0.004 | 0.286 $\pm$ 0.048        | 0.427 $\pm$ 0.072 | 0.159 $\pm$ 0.011 | 0.298 $\pm$ 0.010 | 0.583 $\pm$ 0.063 | 0.895 $\pm$ 0.145 |
| IS95               | <b>0.162</b> $\pm$ 0.008 | 0.606 $\pm$ 0.107        | 1.520 $\pm$ 0.262 | 0.384 $\pm$ 0.033 | 0.600 $\pm$ 0.022 | 1.908 $\pm$ 0.274 | 3.060 $\pm$ 0.588 |
| RE                 | 0.110 $\pm$ 0.006        | <b>0.069</b> $\pm$ 0.004 | 0.157 $\pm$ 0.006 | 0.185 $\pm$ 0.006 | 0.117 $\pm$ 0.003 | 0.302 $\pm$ 0.003 | 0.185 $\pm$ 0.002 |
| (c) Half-Cheetah   |                          |                          |                   |                   |                   |                   |                   |
|                    | GP-PI-K                  | GP                       | GP-NN-K           | GP-PI-M           | GP-NN-M           | DE                |                   |
| CRPS               | <b>0.035</b> $\pm$ 0.006 | 0.125 $\pm$ 0.009        | 0.084 $\pm$ 0.024 | 0.173 $\pm$ 0.012 | 0.306 $\pm$ 0.022 | 0.517 $\pm$ 0.029 |                   |
| IS68               | <b>0.257</b> $\pm$ 0.057 | 0.726 $\pm$ 0.049        | 0.523 $\pm$ 0.114 | 0.942 $\pm$ 0.064 | 1.644 $\pm$ 0.094 | 2.176 $\pm$ 0.118 |                   |
| IS95               | <b>1.150</b> $\pm$ 0.098 | 2.256 $\pm$ 0.101        | 1.937 $\pm$ 0.194 | 2.814 $\pm$ 0.147 | 3.769 $\pm$ 0.200 | 4.652 $\pm$ 0.211 |                   |
| RE                 | 0.072 $\pm$ 0.004        | <b>0.060</b> $\pm$ 0.004 | 0.103 $\pm$ 0.005 | 0.101 $\pm$ 0.003 | 0.287 $\pm$ 0.008 | 0.197 $\pm$ 0.002 |                   |
| (d) 7-DOF Reacher  |                          |                          |                   |                   |                   |                   |                   |
|                    | GP-PI-K                  | GP                       | GP-NN-K           | GP-PI-M           | GP-NN-M           | DE                |                   |
| CRPS               | <b>0.018</b> $\pm$ 0.002 | 0.031 $\pm$ 0.004        | 0.027 $\pm$ 0.005 | 0.049 $\pm$ 0.003 | 0.085 $\pm$ 0.006 | 0.109 $\pm$ 0.006 |                   |
| IS68               | <b>0.129</b> $\pm$ 0.008 | 0.337 $\pm$ 0.013        | 0.226 $\pm$ 0.012 | 0.490 $\pm$ 0.010 | 0.516 $\pm$ 0.017 | 0.632 $\pm$ 0.024 |                   |
| IS95               | <b>0.382</b> $\pm$ 0.027 | 0.685 $\pm$ 0.040        | 0.524 $\pm$ 0.050 | 1.091 $\pm$ 0.054 | 2.552 $\pm$ 0.092 | 3.098 $\pm$ 0.137 |                   |
| RE                 | 0.085 $\pm$ 0.002        | <b>0.056</b> $\pm$ 0.002 | 0.115 $\pm$ 0.003 | 0.154 $\pm$ 0.002 | 0.291 $\pm$ 0.004 | 0.229 $\pm$ 0.004 |                   |



**Fig. 4.** Log predictive stds for the low-dimensional systems. Top: 2-link Reacher over the state space  $\theta_1, \theta_2 \in [-\pi, \pi]$  with 50 training samples. Bottom: 3-link Reacher over the state space  $\theta_1, \theta_2 \in [-\pi, \pi]$  with 250 training samples. Blue dots indicate training data locations.

the imperfect prior model is insufficient for long-horizon prediction without additional probabilistic correction.

#### 5.1.4. Out-of-distribution prediction performance

The one-step prediction accuracy, uncertainty estimation and calibration, and multi-step prediction accuracy results reported in the previous subsections are evaluated on the original test sets, whose data-collection distributions are consistent with those of the training data. In this subsection, we refer to these original test settings as the in-distribution (ID) settings. To further evaluate extrapolation ability under distribution shift, we construct additional out-of-distribution (OOD) test settings for all four systems. For the 2-link Reacher, 3-link Reacher, and 7-DOF Reacher systems, the OOD test trajectories are generated using unseen action ranges. Specifically, the action sampling range used for OOD testing is enlarged relative to that used for training-data collection, which induces trajectories in state-action regions outside the original data-collection distribution. For the Half-Cheetah system, we construct OOD test data using contact-rich rollouts that contain repeated flipping and strong ground impacts.

Table 12 reports the normalized 1-step prediction RMSE and the normalized final 20-step prediction RMSE under the OOD setting. Since the purpose of this OOD evaluation is to examine whether prior information improves extrapolation robustness under distribution shift, we compare GP-PI-K with the applicable prior-informed baselines and the strongest non-prior baseline on each system, rather than repeating the full baseline comparison. Specifically, for the 2-link and 3-link Reacher systems, we compare GP-PI-K with GP-PI-M, AutoIP, and GP, where GP is the strongest non-prior baseline. For Half-Cheetah and 7-DOF Reacher, we

compare GP-PI-K with GP-PI-M and GP-NN-K, where GP-NN-K is the strongest non-prior baseline. Across all four systems and both prediction horizons, GP-PI-K achieves the lowest OOD prediction errors, indicating stronger prediction performance under distribution shift.

To further quantify robustness under distribution shift, we compute the degradation factor from the ID setting to the OOD setting. For each prediction horizon, the degradation factor is defined as

$$DF = \frac{nRMSE_{OOD}}{nRMSE_{ID}}, \quad (63)$$

where  $nRMSE_{ID}$  denotes the normalized RMSE on the original ID test set for the corresponding prediction horizon, and  $nRMSE_{OOD}$  denotes the normalized RMSE on the newly constructed OOD test set. A smaller degradation factor indicates that the model is less affected by the distribution shift and therefore has stronger extrapolation robustness.

As shown in Fig. 6, GP-PI-K consistently exhibits smaller degradation factors than the strongest non-prior baseline for both 1-step and final 20-step prediction across all four systems. This indicates that incorporating prior knowledge through the prior-informed kernel reduces the sensitivity of the learned dynamics model to OOD action ranges and contact-rich rollouts, leading to stronger extrapolation robustness.

#### 5.2. Performance on downstream tasks

We assess the utility of the learned models on downstream tasks, including AL and MBRL, both of which are highly sensitive to the quality of predictive uncertainty. Table 13 summarizes the planning horizons used for AL and MBRL.

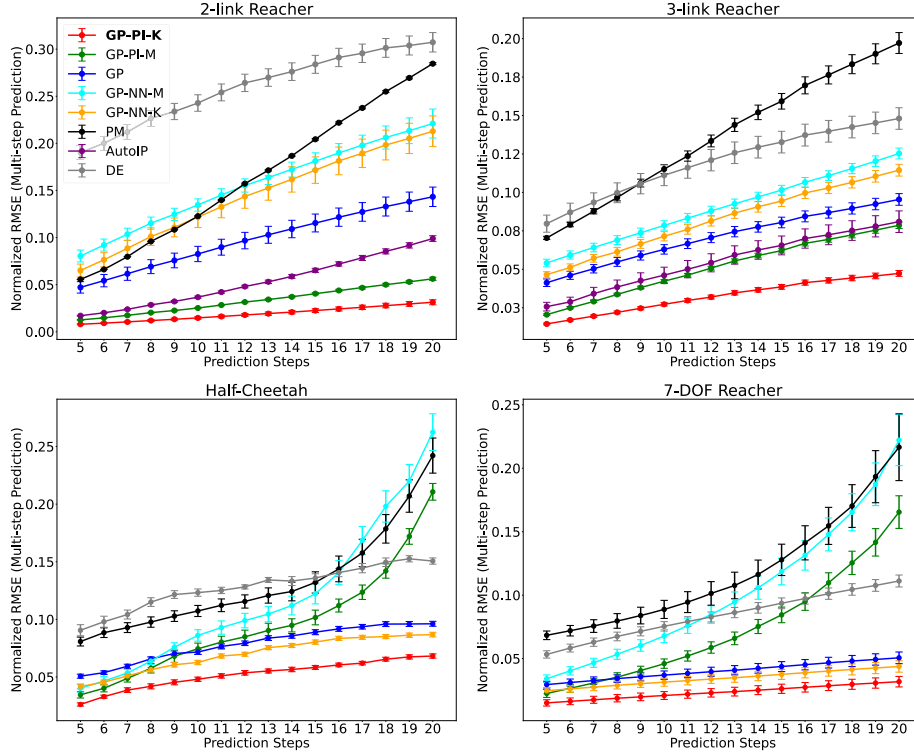


Fig. 5. Normalized RMSEs for multi-step prediction.

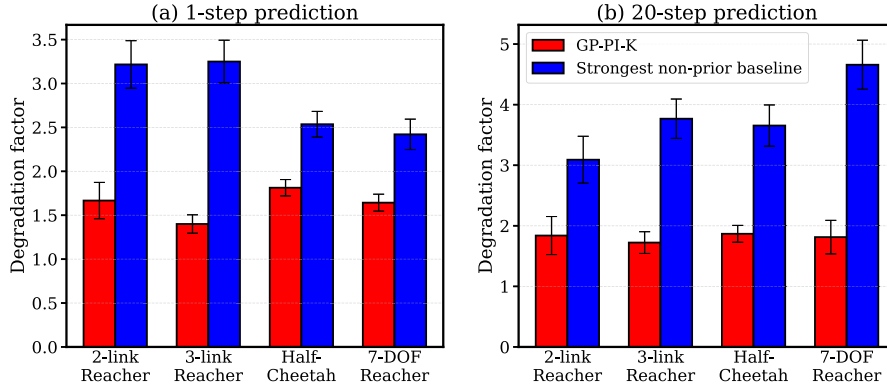


Fig. 6. Degradation factors from the in-distribution (ID) test setting to the out-of-distribution (OOD) test setting. For each random seed, the degradation factor is defined as  $DF = nRMSE_{OOD}/nRMSE_{ID}$ . Smaller values indicate less performance degradation under distribution shift and hence stronger extrapolation robustness. For the 2-link and 3-link Reacher systems, the strongest non-prior baseline is GP. For Half-Cheetah and 7-DOF Reacher, the strongest non-prior baseline is GP-NN-K. Error bars show 95% confidence intervals over 20 random seeds.

### 5.2.1. Performance on active learning

Figs. 7–10 show the normalized RMSEs for 1-step, 5-step, 10-step, and 20-step predictions over AL iterations. For the 2-link Reacher and 3-link Reacher systems, the RMSEs of GP-NN-M, GP-NN-K, and DE are substantially higher than those of the other models. To improve clarity and better highlight the comparison between GP-PI-K, GP-PI-M, and AutoIP, we omit the results of GP-NN-M, GP-NN-K, and DE in these two systems. For the Half-Cheetah and 7-DOF Reacher systems, we also exclude the error bars of GP-PI-M, GP-NN-M, and DE in the 20-step prediction plots due to their excessively large CIs.

Active learning relies on predictive uncertainty to identify informative samples. Figs. 7 to 10 show that the normalized RMSEs of most models decrease as the AL iterations proceed, indicating that actively selected samples improve the learned dynamics models. Across all four systems and all prediction horizons, GP-PI-K consistently achieves the lowest RMSEs and the fastest convergence. This demonstrates that GP-

PI-K is the most effective model for guiding informative sample acquisition in active learning.

For the 2-link Reacher and 3-link Reacher systems, AutoIP performs better than GP throughout the AL process, which suggests that incorporating ODE-based prior knowledge improves sample efficiency in these simple systems. However, AutoIP remains consistently worse than GP-PI-K and GP-PI-M, which is consistent with the one-step and multi-step prediction results. For the Half-Cheetah and 7-DOF Reacher systems, GP-NN-K is the strongest competing baseline and generally performs better than GP, GP-PI-M, GP-NN-M, and DE, although it is still consistently inferior to GP-PI-K.

By contrast, GP-PI-M and GP-NN-M exhibit substantially slower improvement, especially on the complex systems and at longer prediction horizons. This is consistent with their weaker uncertainty calibration and poorer long-horizon prediction performance observed earlier. In particular, for the Half-Cheetah and 7-DOF Reacher systems, the 10-step

**Table 12**

Out-of-distribution prediction performance averaged over output dimensions (mean  $\pm$  95% CI over 20 random seeds). The row “1-step nRMSE” denotes the normalized RMSE for one-step prediction, and “20-step nRMSE” denotes the normalized RMSE at the final 20-step prediction horizon. The comparison includes the applicable prior-informed baselines and the strongest non-prior baseline on each system. Lower values are better.

| (a) 2-link Reacher |                          |                   |                   |                   |
|--------------------|--------------------------|-------------------|-------------------|-------------------|
|                    | GP-PI-K                  | GP-PI-M           | GP                | AutoIP            |
| 1-step nRMSE       | <b>0.005</b> $\pm$ 0.001 | 0.006 $\pm$ 0.001 | 0.074 $\pm$ 0.006 | 0.012 $\pm$ 0.001 |
| 20-step nRMSE      | <b>0.057</b> $\pm$ 0.005 | 0.091 $\pm$ 0.006 | 0.442 $\pm$ 0.027 | 0.147 $\pm$ 0.012 |
| (b) 3-link Reacher |                          |                   |                   |                   |
|                    | GP-PI-K                  | GP-PI-M           | GP                | AutoIP            |
| 1-step nRMSE       | <b>0.007</b> $\pm$ 0.001 | 0.008 $\pm$ 0.001 | 0.039 $\pm$ 0.003 | 0.013 $\pm$ 0.001 |
| 20-step nRMSE      | <b>0.081</b> $\pm$ 0.003 | 0.132 $\pm$ 0.004 | 0.358 $\pm$ 0.012 | 0.148 $\pm$ 0.010 |
| (c) Half-Cheetah   |                          |                   |                   |                   |
|                    | GP-PI-K                  | GP-PI-M           | GP-NN-K           |                   |
| 1-step nRMSE       | <b>0.029</b> $\pm$ 0.001 | 0.031 $\pm$ 0.001 | 0.071 $\pm$ 0.004 |                   |
| 20-step nRMSE      | <b>0.127</b> $\pm$ 0.005 | 0.393 $\pm$ 0.040 | 0.318 $\pm$ 0.030 |                   |
| (d) 7-DOF Reacher  |                          |                   |                   |                   |
|                    | GP-PI-K                  | GP-PI-M           | GP-NN-K           |                   |
| 1-step nRMSE       | <b>0.023</b> $\pm$ 0.001 | 0.025 $\pm$ 0.002 | 0.046 $\pm$ 0.002 |                   |
| 20-step nRMSE      | <b>0.058</b> $\pm$ 0.008 | 0.292 $\pm$ 0.033 | 0.205 $\pm$ 0.032 |                   |

**Table 13**

Planning horizons of AL and MBRL.

|                  | 2-link Reacher | 3-link Reacher | Half-Cheetah | 7-DOF Reacher |
|------------------|----------------|----------------|--------------|---------------|
| AL horizon $T'$  | 20             | 30             | 200          | 100           |
| MBRL horizon $T$ | 50             | 50             | 1,000        | 200           |

and 20-step prediction errors of GP-PI-M, GP-NN-M, and DE remain relatively high even after many AL iterations. These results indicate that strong active-learning performance requires not only accurate one-step prediction, but also reliable uncertainty behavior and robustness in recursive prediction.

Overall, these findings highlight the importance of reliable uncertainty quantification for efficient exploration and sample-efficient model learning. GP-PI-K improves model accuracy while preserving useful GP-style uncertainty behavior, which makes it particularly effective for active learning.

### 5.2.2. Performance on model-based reinforcement learning

Fig. 11 presents the MBRL performance in terms of episode reward over training iterations. We include model-free reinforcement learning algorithms, namely Soft Actor-Critic (SAC) and Trust Region Policy Optimization (TRPO), as reference baselines. These model-free methods serve as upper-reference curves because they are not affected by model errors, although they require substantially more environment interactions.

Across all four systems, GP-PI-K consistently achieves the best control performance. This result is consistent with its strong prediction accuracy and its favorable uncertainty behavior observed in the previous subsections. In particular, GP-PI-K converges faster and attains higher final rewards than all other model-based baselines.

For the 2-link Reacher and 3-link Reacher systems, GP-PI-M also performs well and remains the second-best method in most cases. AutoIP outperforms GP in these simple systems. For the Half-Cheetah and 7-DOF Reacher systems, the advantage of GP-PI-K becomes even more pronounced. GP-NN-K is the strongest competing baseline on these complex systems and performs better than GP-NN-M and DE, but it still lags behind GP-PI-K. By contrast, GP-PI-M yields substantially worse rewards than GP-PI-K on these systems, indicating that a mean-function-based

**Table 14**

Paired Wilcoxon signed-rank test results for GP-PI-K against the strongest baseline on each system. A  $p$ -value smaller than 0.05 indicates statistical significance.

| System         | MNLP                 | RMSE <sub>20-step</sub> <sup>Norm</sup> | AL (final 10-step)   | MBRL (final)         |
|----------------|----------------------|-----------------------------------------|----------------------|----------------------|
| 2-link Reacher | $9.5 \times 10^{-7}$ | $9.5 \times 10^{-7}$                    | $9.5 \times 10^{-7}$ | $1.9 \times 10^{-6}$ |
| 3-link Reacher | $3.1 \times 10^{-5}$ | $9.5 \times 10^{-7}$                    | $9.5 \times 10^{-7}$ | $9.5 \times 10^{-7}$ |
| Half-Cheetah   | $9.5 \times 10^{-7}$ | $9.5 \times 10^{-7}$                    | $9.5 \times 10^{-7}$ | $1.9 \times 10^{-6}$ |
| 7-DOF Reacher  | $9.5 \times 10^{-7}$ | $2.8 \times 10^{-3}$                    | $9.5 \times 10^{-7}$ | $2.0 \times 10^{-4}$ |

**Table 15**

Paired Wilcoxon signed-rank test results comparing GP-PI-K and GP in terms of absolute empirical coverage error. For each random seed, the test statistics are computed from  $|\text{cov68} - 0.68|$  and  $|\text{cov95} - 0.95|$ . The entries report the resulting  $p$ -values, not the coverage errors themselves. A  $p$ -value larger than 0.05 indicates that no statistically significant difference is detected.

| System         | $ \text{cov68} - 0.68 $ | $ \text{cov95} - 0.95 $ |
|----------------|-------------------------|-------------------------|
| 2-link Reacher | 0.165                   | 0.452                   |
| 3-link Reacher | 0.058                   | 0.388                   |
| Half-Cheetah   | 0.177                   | 0.349                   |
| 7-DOF Reacher  | 0.083                   | 0.202                   |

incorporation of prior knowledge is less robust when the prior model is imperfect and the control task requires long-horizon recursive prediction.

Overall, these results show that strong MBRL performance requires not only accurate dynamics prediction but also reliable uncertainty behavior during planning. GP-PI-K provides the most effective balance of these properties, which leads to the best control performance across all evaluated systems.

### 5.3. Statistical significance tests

To further assess whether the observed improvements of GP-PI-K are statistically significant, we conduct paired Wilcoxon signed-rank tests against the strongest baseline on each system. For MNLP, we first average the MNLP values over all output dimensions for each random seed, and then perform the paired test on the resulting overall MNLP values. For multi-step prediction, we use the normalized RMSE at the final 20-step prediction horizon, i.e.,  $\text{RMSE}_{20\text{-step}}^{\text{Norm}}$ . For AL, we use the 10-step prediction result at the final iteration. For MBRL, we use the control performance at the final iteration. Table 14 reports the resulting  $p$ -values. In all cases, the  $p$ -values are below 0.05, indicating that the improvements of GP-PI-K over the strongest competing baseline are statistically significant.

For empirical coverage, larger values are not necessarily better; values closer to the nominal levels indicate better calibration. Therefore, we conduct a separate paired test for calibration based on absolute empirical coverage errors. Specifically, for each random seed, we compute  $|\text{cov68} - 0.68|$  and  $|\text{cov95} - 0.95|$  for GP-PI-K and GP, respectively. We then apply paired Wilcoxon signed-rank tests to the paired absolute coverage errors of GP-PI-K and GP under the same random seeds. The null hypothesis is that the median paired difference in absolute coverage error between GP-PI-K and GP is zero. Table 15 reports the resulting  $p$ -values. Since all  $p$ -values are larger than 0.05, no statistically significant difference is detected between GP-PI-K and GP in terms of absolute empirical coverage error. This result supports the revised claim that GP-PI-K largely preserves the favorable calibration behavior of GP, while substantially improving predictive accuracy and overall probabilistic prediction quality.

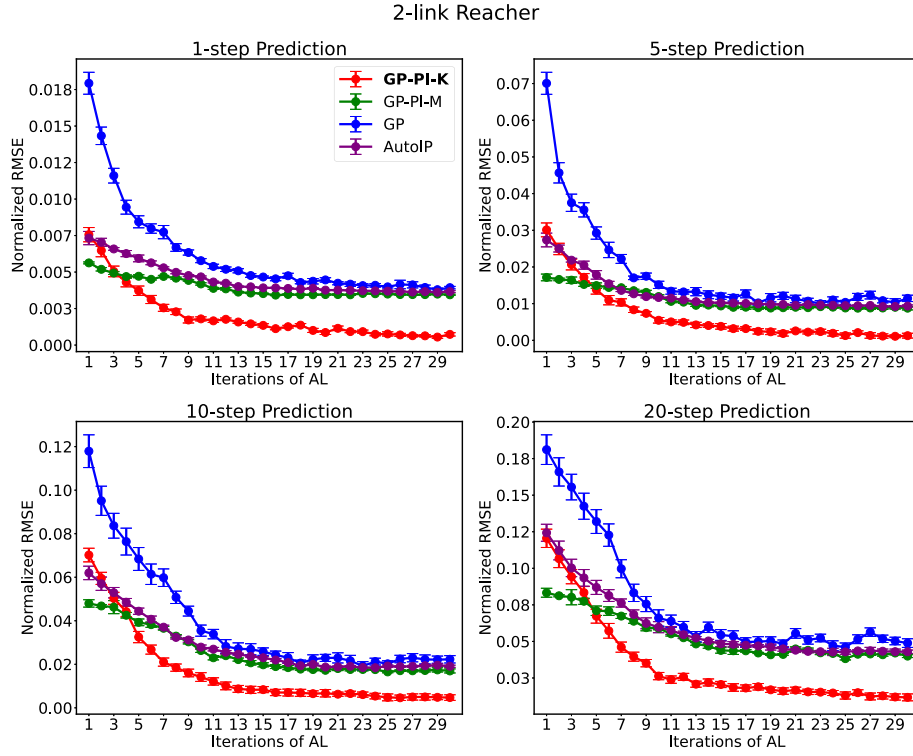


Fig. 7. 2-link Reacher: AL results (normalized RMSEs vs. iteration).

Table 16

2-link Reacher: ablation results for the residual kernel and learned prior parameters. Here, “w/o res.” denotes the variant of our proposed GP-PI-K without the residual kernel, and “fixed prior” denotes the variant of GP-PI-K in which the prior parameters are fixed rather than jointly learned with the GP parameters. Reported values are means over 20 random seeds. Due to space limitations, 95% CIs are not shown.

|                  | RMSE         |          |             | MNLP          |          |             |
|------------------|--------------|----------|-------------|---------------|----------|-------------|
|                  | GP-PI-K      | w/o res. | fixed prior | GP-PI-K       | w/o res. | fixed prior |
| $\cos \theta_1$  | <b>0.011</b> | 0.033    | 0.013       | <b>-3.103</b> | -1.526   | -2.562      |
| $\cos \theta_2$  | <b>0.006</b> | 0.026    | 0.011       | <b>-3.513</b> | -1.613   | -2.820      |
| $\sin \theta_1$  | <b>0.008</b> | 0.037    | 0.011       | <b>-3.202</b> | -1.892   | -2.928      |
| $\sin \theta_2$  | <b>0.003</b> | 0.029    | 0.008       | <b>-3.599</b> | -1.650   | -2.846      |
| $\dot{\theta}_1$ | <b>0.013</b> | 0.062    | 0.018       | <b>-2.638</b> | -1.102   | -2.169      |
| $\dot{\theta}_2$ | <b>0.012</b> | 0.039    | 0.016       | <b>-2.744</b> | -1.280   | -2.371      |

Table 17

3-link Reacher: ablation results for the residual kernel and learned prior parameters.

|                  | RMSE         |          |             | MNLP          |          |             |
|------------------|--------------|----------|-------------|---------------|----------|-------------|
|                  | GP-PI-K      | w/o res. | fixed prior | GP-PI-K       | w/o res. | fixed prior |
| $\cos \theta_1$  | <b>0.010</b> | 0.023    | 0.016       | <b>-3.415</b> | -1.920   | -2.862      |
| $\cos \theta_2$  | <b>0.009</b> | 0.021    | 0.015       | <b>-3.215</b> | -1.864   | -2.783      |
| $\cos \theta_3$  | <b>0.011</b> | 0.025    | 0.016       | <b>-3.358</b> | -1.767   | -2.666      |
| $\sin \theta_1$  | <b>0.012</b> | 0.024    | 0.017       | <b>-3.429</b> | -1.966   | -2.694      |
| $\sin \theta_2$  | <b>0.012</b> | 0.027    | 0.018       | <b>-3.433</b> | -1.666   | -2.738      |
| $\sin \theta_3$  | <b>0.009</b> | 0.021    | 0.017       | <b>-3.422</b> | -1.776   | -2.752      |
| $\dot{\theta}_1$ | <b>0.014</b> | 0.180    | 0.022       | <b>-3.068</b> | -0.502   | -2.516      |
| $\dot{\theta}_2$ | <b>0.020</b> | 0.140    | 0.032       | <b>-2.913</b> | -0.713   | -2.496      |
| $\dot{\theta}_3$ | <b>0.037</b> | 0.191    | 0.048       | <b>-2.801</b> | -0.460   | -2.587      |

#### 5.4. Ablation studies

For brevity, the ablation tables report only the mean results over 20 random seeds; the same trends are consistently observed across seeds.

##### 5.4.1. Effect of the residual kernel and learned prior parameters

We first investigate two design choices in GP-PI-K: the residual kernel and the learning of prior-model parameters. For the residual-kernel ablation, we compare the full GP-PI-K with a variant without the residual kernel. For the prior-parameter ablation, we compare the full GP-PI-K with a variant in which the prior parameters are fixed to their nominal values rather than jointly learned with the GP parameters. Since the 2-link Reacher and 3-link Reacher systems use physical prior models with learnable physical parameters, both ablations are conducted on these two systems. For the Half-Cheetah and 7-DOF Reacher systems, the prior model is a fixed surrogate NN, and therefore only the residual-kernel ablation is applicable. Tables 16–18 report the corresponding one-step RMSE and MNLP results.

The results lead to two clear observations. First, the residual kernel is essential. Removing it causes a substantial degradation in both prediction accuracy and uncertainty quality across all four systems. This confirms that the residual kernel plays a critical role in compensating for the mismatch between the prior model and the true system. Without this correction mechanism, the prior-model-informed kernel alone is insufficient to capture the remaining discrepancy.

Second, fixing the prior parameters to their nominal values generally leads to slightly worse results than jointly learning them with the GP parameters. This trend is observed consistently on both the 2-link Reacher and 3-link Reacher systems. The performance gap is not as dramatic as that caused by removing the residual kernel, but it nevertheless indicates that jointly adapting the prior parameters is beneficial. At the same time, the fact that the full GP-PI-K consistently outperforms the fixed-prior variant suggests that learning the prior parameters does not simply distort the prior model to artificially improve the likelihood; instead, it provides a meaningful refinement of the prior model that improves both prediction accuracy and uncertainty estimation.

##### 5.4.2. Sensitivity to prior quality / simulator mismatch

We next examine the robustness of GP-PI-K under different levels of prior-model mismatch. We consider three prior-gap settings: small,



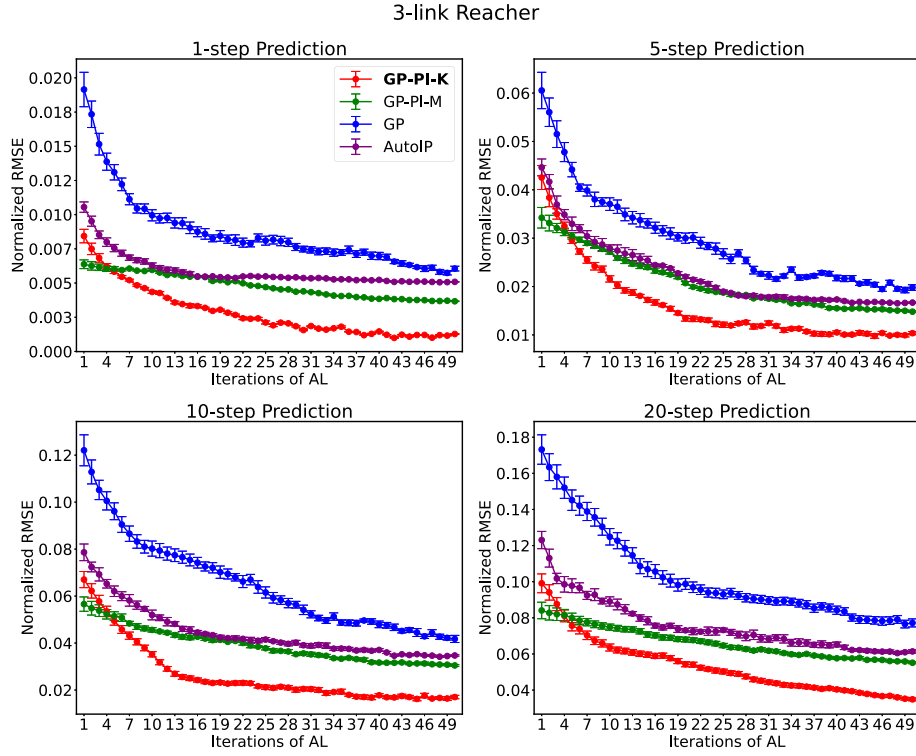


Fig. 8. 3-link Reacher: AL results (normalized RMSEs vs. iteration).

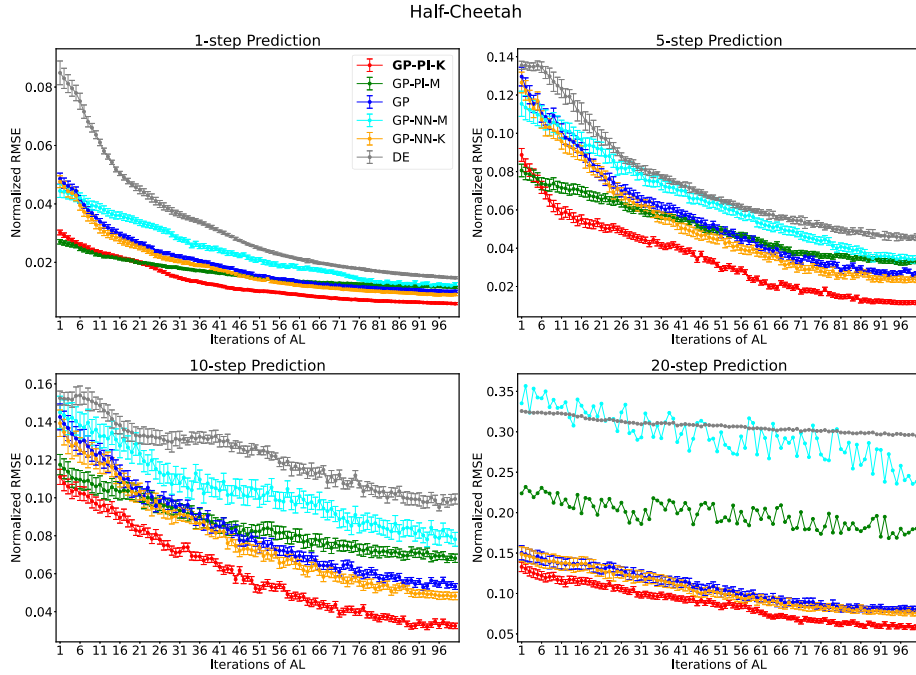


Fig. 9. Half-Cheetah: AL results (normalized RMSEs vs. iteration).

medium, and large, where the medium-gap setting corresponds to the main experimental setting used in the previous subsections.

For the 2-link Reacher and 3-link Reacher systems, the prior model is the ideal ODE that ignores friction and damping, while the true system is the MuJoCo simulator. We control the prior gap by varying the friction and damping coefficients in the true system. In this ablation, GP-PI-K keeps its original form and still uses learned prior parameters. We compare the three prior-informed methods, GP-PI-K, GP-PI-M, and

AutoIP, together with the strongest non-prior baseline, GP. The results are reported in [Tables 19](#) and [20](#).

For the Half-Cheetah and 7-DOF Reacher systems, both the prior and the true system are based on MuJoCo. We control the prior gap by varying the morphology difference between the prior system and the true system. For example, in the Half-Cheetah system, we vary the torso length of the prior simulator to construct different gap levels: a 1.1× torso length corresponds to the small-gap setting, a 1.5× torso length

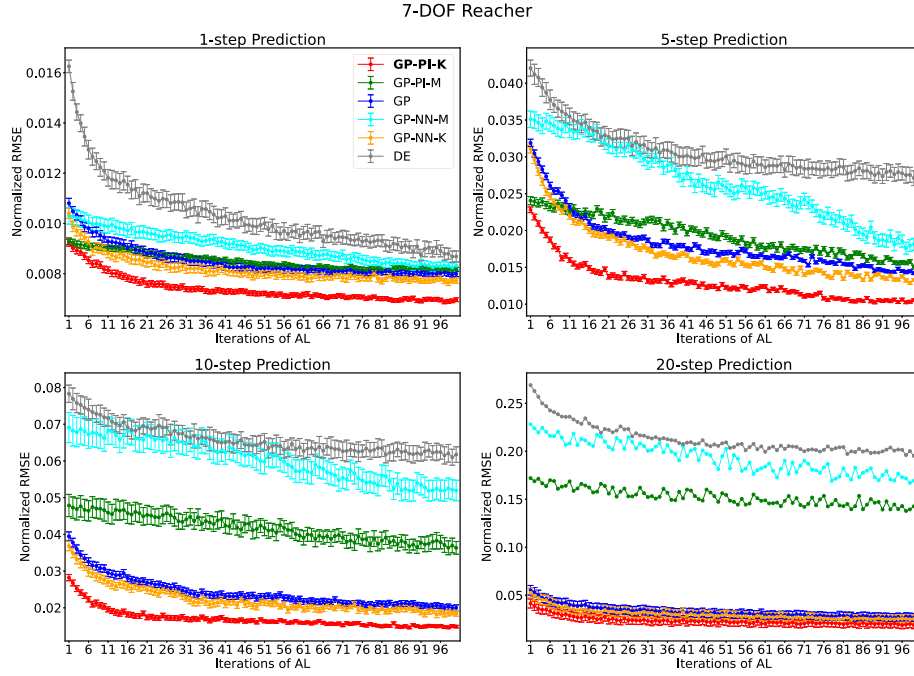


Fig. 10. 7-DOF Reacher: AL results (normalized RMSEs vs. iteration).

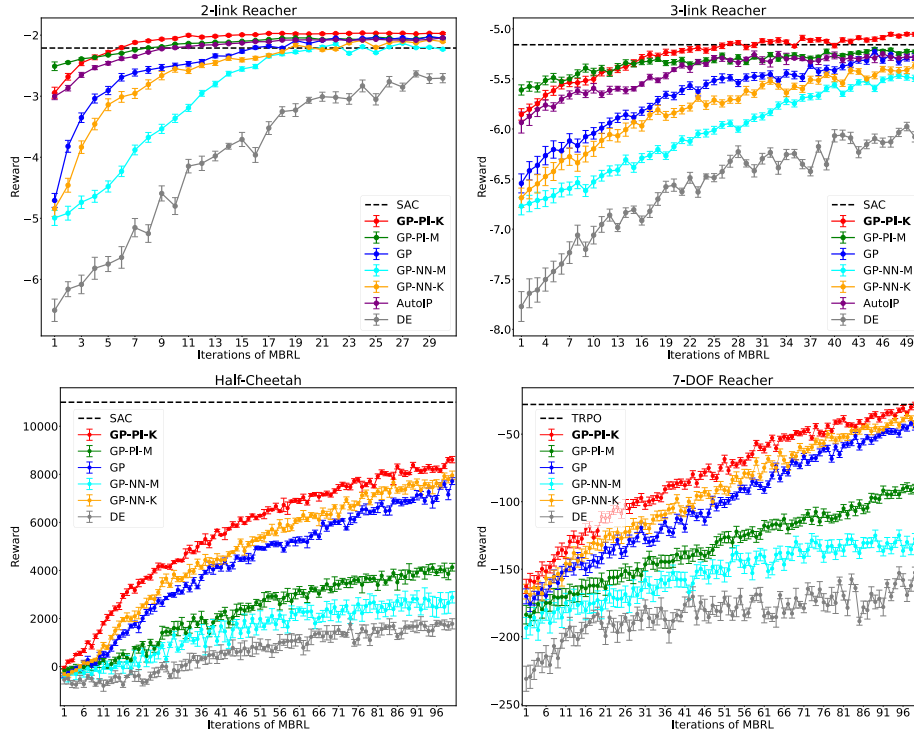


Fig. 11. Performance of MBRL (Episode reward vs. iteration). For the 7-DOF Reacher task, TRPO is adopted instead of SAC, which fails to solve the task.

corresponds to the medium-gap setting, and a  $2.0\times$  torso length corresponds to the large-gap setting. Similarly, in the 7-DOF Reacher system, the gap is controlled by scaling the upper-arm and forearm lengths of the prior simulator. For each gap setting, a surrogate NN is first trained on prior-generated data and then fixed inside GP-PI-K. We compare the two prior-informed methods, GP-PI-K and GP-PI-M, together with the strongest non-prior baseline, GP-NN-K. Since the true system is unchanged, the GP-NN-K results are identical across the three gap settings. The results are reported in Tables 21 and 22.

Several conclusions can be drawn from the small, medium, and large prior-gap settings. First, when the prior gap is smaller, prior-informed methods generally benefit more from the prior model. This trend is consistently observed for GP-PI-K, GP-PI-M, and AutoIP on the 2-link and 3-link Reacher systems, and for GP-PI-K and GP-PI-M on the Half-Cheetah and 7-DOF Reacher systems. Second, as the prior gap increases, the performance of prior-informed methods degrades. However, the degradation of GP-PI-K is typically more gradual than that of the other prior-informed baselines, indicating stronger robustness to moderate

**Table 18**

Half-Cheetah and 7-DOF Reacher: ablation results for the residual kernel.

| (a) Half-Cheetah                  |              |          |               |          |
|-----------------------------------|--------------|----------|---------------|----------|
|                                   | RMSE         |          | MNLP          |          |
|                                   | GP-PI-K      | w/o res. | GP-PI-K       | w/o res. |
| $\frac{x_{\text{res}} - x_t}{dt}$ | <b>0.206</b> | 0.403    | <b>-0.885</b> | -0.101   |
| $z$                               | <b>0.007</b> | 0.018    | <b>-3.676</b> | -1.523   |
| $\cos \theta_1$                   | <b>0.013</b> | 0.023    | <b>-3.760</b> | -1.821   |
| $\cos \theta_2$                   | <b>0.008</b> | 0.020    | <b>-3.609</b> | -1.792   |
| $\cos \theta_3$                   | <b>0.008</b> | 0.019    | <b>-3.425</b> | -1.482   |
| $\cos \theta_4$                   | <b>0.006</b> | 0.013    | <b>-3.965</b> | -1.742   |
| $\cos \theta_5$                   | <b>0.013</b> | 0.023    | <b>-3.033</b> | -1.686   |
| $\cos \theta_6$                   | <b>0.012</b> | 0.027    | <b>-3.146</b> | -1.718   |
| $\cos \theta_7$                   | <b>0.007</b> | 0.021    | <b>-3.582</b> | -1.777   |
| $\sin \theta_1$                   | <b>0.014</b> | 0.027    | <b>-3.099</b> | -1.908   |
| $\sin \theta_2$                   | <b>0.017</b> | 0.039    | <b>-2.757</b> | -1.395   |
| $\sin \theta_3$                   | <b>0.018</b> | 0.045    | <b>-2.663</b> | -1.615   |
| $\sin \theta_4$                   | <b>0.019</b> | 0.042    | <b>-2.683</b> | -1.741   |
| $\sin \theta_5$                   | <b>0.028</b> | 0.051    | <b>-2.287</b> | -1.502   |
| $\sin \theta_6$                   | <b>0.027</b> | 0.048    | <b>-2.285</b> | -1.030   |
| $\sin \theta_7$                   | <b>0.024</b> | 0.047    | <b>-2.338</b> | -1.208   |
| $\dot{x}$                         | <b>0.219</b> | 0.481    | <b>-0.438</b> | 0.377    |
| $\dot{z}$                         | <b>0.182</b> | 0.392    | <b>-0.376</b> | 0.596    |
| $\dot{\theta}_1$                  | <b>0.387</b> | 0.631    | <b>0.110</b>  | 0.884    |
| $\dot{\theta}_2$                  | <b>0.735</b> | 1.234    | <b>0.529</b>  | 1.235    |
| $\dot{\theta}_3$                  | <b>1.027</b> | 1.512    | <b>0.684</b>  | 1.419    |
| $\dot{\theta}_4$                  | <b>1.002</b> | 1.623    | <b>0.691</b>  | 2.216    |
| $\dot{\theta}_5$                  | <b>1.003</b> | 2.631    | <b>0.693</b>  | 2.368    |
| $\dot{\theta}_6$                  | <b>1.397</b> | 2.548    | <b>0.831</b>  | 2.786    |
| $\dot{\theta}_7$                  | <b>1.065</b> | 1.402    | <b>0.735</b>  | 2.500    |
| (b) 7-DOF Reacher                 |              |          |               |          |
|                                   | RMSE         |          | MNLP          |          |
|                                   | GP-PI-K      | w/o res. | GP-PI-K       | w/o res. |
| $\cos \theta_1$                   | <b>0.001</b> | 0.006    | <b>-6.484</b> | -2.156   |
| $\cos \theta_2$                   | <b>0.006</b> | 0.015    | <b>-6.335</b> | -3.331   |
| $\cos \theta_3$                   | <b>0.011</b> | 0.036    | <b>-3.807</b> | -1.225   |
| $\cos \theta_4$                   | <b>0.014</b> | 0.038    | <b>-3.358</b> | -1.133   |
| $\cos \theta_5$                   | <b>0.008</b> | 0.026    | <b>-3.392</b> | -1.041   |
| $\cos \theta_6$                   | <b>0.005</b> | 0.017    | <b>-3.525</b> | -1.298   |
| $\cos \theta_7$                   | <b>0.008</b> | 0.063    | <b>-3.371</b> | -1.164   |
| $\sin \theta_1$                   | <b>0.005</b> | 0.011    | <b>-5.024</b> | -2.725   |
| $\sin \theta_2$                   | <b>0.017</b> | 0.023    | <b>-4.816</b> | -2.124   |
| $\sin \theta_3$                   | <b>0.006</b> | 0.030    | <b>-3.940</b> | -1.612   |
| $\sin \theta_4$                   | <b>0.005</b> | 0.019    | <b>-4.219</b> | -1.723   |
| $\sin \theta_5$                   | <b>0.011</b> | 0.042    | <b>-3.560</b> | -1.542   |
| $\sin \theta_6$                   | <b>0.004</b> | 0.029    | <b>-3.946</b> | -1.875   |
| $\sin \theta_7$                   | <b>0.010</b> | 0.044    | <b>-3.410</b> | -1.525   |
| $\dot{\theta}_1$                  | <b>0.009</b> | 0.017    | <b>-3.342</b> | -1.126   |
| $\dot{\theta}_2$                  | <b>0.011</b> | 0.036    | <b>-3.207</b> | -1.773   |
| $\dot{\theta}_3$                  | <b>0.066</b> | 0.140    | <b>-1.256</b> | 1.030    |
| $\dot{\theta}_4$                  | <b>0.053</b> | 0.117    | <b>-1.327</b> | 1.233    |
| $\dot{\theta}_5$                  | <b>0.031</b> | 0.101    | <b>-2.069</b> | 1.343    |
| $\dot{\theta}_6$                  | <b>0.135</b> | 0.251    | <b>-0.651</b> | 1.548    |
| $\dot{\theta}_7$                  | <b>0.173</b> | 0.363    | <b>-1.494</b> | 1.723    |

prior mismatch. Third, GP-PI-K remains the best prior-informed method across different systems and different gap settings. Most importantly, under the small, medium, and large prior-gap settings considered in Tables 19–22, GP-PI-K consistently outperforms the strongest non-prior baseline, which demonstrates that the proposed method remains effective when the prior model is imperfect but still informative.

To further examine the performance decline of GP-PI-K when the prior model becomes severely inaccurate, we extend the prior-model mismatch levels from small, medium, and large to five levels: small, medium, large, severe, and misleading. The small-to-large settings correspond to imperfect but still informative prior models, whereas the severe and misleading settings are designed to examine failure cases where the prior model provides substantially degraded or even misleading similar-

**Table 19**

2-link Reacher: results under different levels of prior-model mismatch. Here, the small-, medium-, and large-gap settings denote increasing mismatch between the true system and the ODE prior model, induced by varying the friction and damping coefficients of the true system. Reported values are means over 20 random seeds. Due to space limitations, 95% CIs are not shown.

| (a) Small gap    |              |              |       |        |               |         |        |        |
|------------------|--------------|--------------|-------|--------|---------------|---------|--------|--------|
|                  | RMSE         |              |       |        | MNLP          |         |        |        |
|                  | GP-PI-K      | GP-PI-M      | GP    | AutoIP | GP-PI-K       | GP-PI-M | GP     | AutoIP |
| $\cos \theta_1$  | 0.010        | <b>0.008</b> | 0.052 | 0.014  | <b>-3.211</b> | -1.708  | -2.312 | -2.885 |
| $\cos \theta_2$  | 0.006        | <b>0.005</b> | 0.051 | 0.010  | <b>-3.617</b> | -1.601  | -2.320 | -2.712 |
| $\sin \theta_1$  | 0.007        | <b>0.006</b> | 0.057 | 0.009  | <b>-3.207</b> | -1.547  | -2.276 | -2.950 |
| $\sin \theta_2$  | <b>0.003</b> | <b>0.003</b> | 0.039 | 0.006  | <b>-3.557</b> | -1.328  | -2.206 | -3.124 |
| $\dot{\theta}_1$ | <b>0.011</b> | 0.012        | 0.051 | 0.015  | <b>-2.712</b> | -1.389  | -2.237 | -2.527 |
| $\dot{\theta}_2$ | 0.011        | <b>0.010</b> | 0.049 | 0.014  | <b>-2.820</b> | -1.344  | -2.219 | -2.486 |
| (b) Medium gap   |              |              |       |        |               |         |        |        |
|                  | RMSE         |              |       |        | MNLP          |         |        |        |
|                  | GP-PI-K      | GP-PI-M      | GP    | AutoIP | GP-PI-K       | GP-PI-M | GP     | AutoIP |
| $\cos \theta_1$  | 0.011        | <b>0.009</b> | 0.060 | 0.016  | <b>-3.103</b> | -1.678  | -2.184 | -2.814 |
| $\cos \theta_2$  | 0.006        | <b>0.005</b> | 0.052 | 0.010  | <b>-3.513</b> | -1.553  | -2.215 | -2.639 |
| $\sin \theta_1$  | 0.008        | <b>0.006</b> | 0.062 | 0.010  | <b>-3.202</b> | -1.467  | -2.142 | -2.842 |
| $\sin \theta_2$  | <b>0.003</b> | 0.004        | 0.046 | 0.006  | <b>-3.599</b> | -1.259  | -2.227 | -2.944 |
| $\dot{\theta}_1$ | <b>0.013</b> | 0.014        | 0.059 | 0.017  | <b>-2.638</b> | -1.369  | -2.077 | -2.421 |
| $\dot{\theta}_2$ | 0.012        | <b>0.010</b> | 0.052 | 0.016  | <b>-2.744</b> | -1.270  | -2.100 | -2.387 |
| (c) Large gap    |              |              |       |        |               |         |        |        |
|                  | RMSE         |              |       |        | MNLP          |         |        |        |
|                  | GP-PI-K      | GP-PI-M      | GP    | AutoIP | GP-PI-K       | GP-PI-M | GP     | AutoIP |
| $\cos \theta_1$  | <b>0.014</b> | 0.015        | 0.062 | 0.023  | <b>-2.950</b> | -1.302  | -2.019 | -2.547 |
| $\cos \theta_2$  | <b>0.007</b> | 0.008        | 0.055 | 0.018  | <b>-3.274</b> | -1.293  | -2.037 | -2.521 |
| $\sin \theta_1$  | <b>0.010</b> | <b>0.010</b> | 0.065 | 0.016  | <b>-2.987</b> | -1.240  | -1.930 | -2.440 |
| $\sin \theta_2$  | <b>0.006</b> | 0.007        | 0.050 | 0.016  | <b>-3.331</b> | -1.029  | -2.146 | -2.674 |
| $\dot{\theta}_1$ | <b>0.015</b> | 0.018        | 0.063 | 0.027  | <b>-2.493</b> | -1.087  | -1.928 | -2.242 |
| $\dot{\theta}_2$ | <b>0.015</b> | <b>0.015</b> | 0.057 | 0.028  | <b>-2.608</b> | -1.120  | -1.984 | -2.198 |

ity information. For the 2-link and 3-link Reacher systems, GP is used as the strongest non-prior baseline. For Half-Cheetah and 7-DOF Reacher, GP-NN-K is used as the strongest non-prior baseline.

As shown in Fig. 12, GP-PI-K benefits substantially from the prior model when the prior is imperfect but still informative. However, its prediction error increases as the prior-model mismatch becomes larger. In the severe setting, the advantage of GP-PI-K becomes smaller; in the misleading setting, this advantage can disappear. In particular, for the Half-Cheetah and 7-DOF Reacher systems, GP-PI-K can become worse than the strongest non-prior baseline when the prior model becomes misleading. This identifies an important failure case: GP-PI-K is not guaranteed to improve prediction performance when the prior model induces misleading similarity in the transformed space. Therefore, GP-PI-K is most effective when the prior model is imperfect but still informative, while severely inaccurate or misleading priors can reduce or remove its advantage.

### 5.5. Computational complexity analysis

We further analyze the computational complexity of GP-PI-K and compare it with that of standard GP models. The main theoretical results are summarized in Table 23. Let  $N$  denote the number of training samples,  $M$  the number of test samples,  $B$  the mini-batch size in SVGP training, and  $M_u$  the number of inducing points. For GP-PI-K, the additional computation mainly comes from evaluating the prior-model transformation and the additive composite kernel. However, this does not change the overall complexity class of the underlying GP framework; instead, it introduces additional constant-factor overhead.

For the 2-link Reacher and 3-link Reacher systems, GP-PI-K uses exact GP inference with an ODE-based prior model. In this case, com-

**Table 20**

3-link Reacher: results under different levels of prior-model mismatch. Here, the small-, medium-, and large-gap settings denote increasing mismatch between the true system and the ODE prior model, induced by varying the friction and damping coefficients of the true system.

| Small gap        |              |              |       |        |               |         |        |        |
|------------------|--------------|--------------|-------|--------|---------------|---------|--------|--------|
|                  | RMSE         |              |       |        | MNLP          |         |        |        |
|                  | GP-PI-K      | GP-PI-M      | GP    | AutoIP | GP-PI-K       | GP-PI-M | GP     | AutoIP |
| $\cos \theta_1$  | 0.009        | <b>0.008</b> | 0.025 | 0.011  | <b>-3.471</b> | -2.317  | -2.601 | -2.929 |
| $\cos \theta_2$  | <b>0.007</b> | <b>0.007</b> | 0.027 | 0.013  | <b>-3.330</b> | -2.470  | -2.587 | -2.845 |
| $\cos \theta_3$  | 0.010        | <b>0.008</b> | 0.024 | 0.014  | <b>-3.352</b> | -2.345  | -2.586 | -3.010 |
| $\sin \theta_1$  | 0.010        | <b>0.008</b> | 0.022 | 0.014  | <b>-3.492</b> | -2.411  | -2.628 | -2.858 |
| $\sin \theta_2$  | 0.009        | <b>0.007</b> | 0.030 | 0.012  | <b>-3.500</b> | -2.389  | -2.507 | -2.705 |
| $\sin \theta_3$  | 0.008        | <b>0.007</b> | 0.018 | 0.013  | <b>-3.537</b> | -2.513  | -2.645 | -3.007 |
| $\dot{\theta}_1$ | <b>0.012</b> | 0.025        | 0.102 | 0.023  | <b>-3.124</b> | -2.124  | -2.036 | -2.786 |
| $\dot{\theta}_2$ | <b>0.019</b> | 0.040        | 0.101 | 0.052  | <b>-3.016</b> | -1.739  | -1.826 | -2.452 |
| $\dot{\theta}_3$ | <b>0.036</b> | 0.042        | 0.068 | 0.047  | <b>-2.875</b> | -2.012  | -2.079 | -2.536 |
| (b) Medium gap   |              |              |       |        |               |         |        |        |
|                  | RMSE         |              |       |        | MNLP          |         |        |        |
|                  | GP-PI-K      | GP-PI-M      | GP    | AutoIP | GP-PI-K       | GP-PI-M | GP     | AutoIP |
| $\cos \theta_1$  | 0.010        | <b>0.009</b> | 0.028 | 0.013  | <b>-3.415</b> | -2.281  | -2.552 | -2.909 |
| $\cos \theta_2$  | 0.009        | <b>0.007</b> | 0.029 | 0.015  | <b>-3.215</b> | -2.484  | -2.541 | -2.792 |
| $\cos \theta_3$  | 0.011        | <b>0.008</b> | 0.025 | 0.014  | <b>-3.358</b> | -2.277  | -2.598 | -2.933 |
| $\sin \theta_1$  | 0.012        | <b>0.008</b> | 0.023 | 0.016  | <b>-3.429</b> | -2.384  | -2.608 | -2.840 |
| $\sin \theta_2$  | 0.012        | <b>0.008</b> | 0.032 | 0.017  | <b>-3.433</b> | -2.290  | -2.430 | -2.622 |
| $\sin \theta_3$  | 0.009        | <b>0.007</b> | 0.019 | 0.015  | <b>-3.422</b> | -2.488  | -2.608 | -2.968 |
| $\dot{\theta}_1$ | <b>0.014</b> | 0.027        | 0.105 | 0.026  | <b>-3.068</b> | -2.111  | -1.962 | -2.729 |
| $\dot{\theta}_2$ | <b>0.020</b> | 0.043        | 0.107 | 0.060  | <b>-2.913</b> | -1.685  | -1.794 | -2.417 |
| $\dot{\theta}_3$ | <b>0.037</b> | 0.046        | 0.071 | 0.053  | <b>-2.801</b> | -1.970  | -2.048 | -2.493 |
| (c) Large gap    |              |              |       |        |               |         |        |        |
|                  | RMSE         |              |       |        | MNLP          |         |        |        |
|                  | GP-PI-K      | GP-PI-M      | GP    | AutoIP | GP-PI-K       | GP-PI-M | GP     | AutoIP |
| $\cos \theta_1$  | <b>0.013</b> | 0.015        | 0.031 | 0.019  | <b>-3.245</b> | -1.940  | -2.501 | -2.729 |
| $\cos \theta_2$  | <b>0.013</b> | 0.014        | 0.034 | 0.023  | <b>-3.029</b> | -2.169  | -2.453 | -2.661 |
| $\cos \theta_3$  | <b>0.015</b> | 0.016        | 0.028 | 0.021  | <b>-3.244</b> | -2.014  | -2.509 | -2.707 |
| $\sin \theta_1$  | 0.016        | <b>0.015</b> | 0.026 | 0.022  | <b>-3.280</b> | -2.055  | -2.520 | -2.613 |
| $\sin \theta_2$  | <b>0.015</b> | 0.016        | 0.038 | 0.025  | <b>-3.347</b> | -1.973  | -2.336 | -2.474 |
| $\sin \theta_3$  | <b>0.012</b> | 0.013        | 0.022 | 0.019  | <b>-3.291</b> | -2.062  | -2.512 | -2.688 |
| $\dot{\theta}_1$ | <b>0.016</b> | 0.037        | 0.119 | 0.041  | <b>-2.842</b> | -1.879  | -1.927 | -2.501 |
| $\dot{\theta}_2$ | <b>0.027</b> | 0.052        | 0.120 | 0.070  | <b>-2.808</b> | -1.548  | -1.678 | -2.122 |
| $\dot{\theta}_3$ | <b>0.040</b> | 0.058        | 0.080 | 0.062  | <b>-2.699</b> | -1.657  | -1.904 | -2.206 |

pared with a standard exact GP, the dominant  $\mathcal{O}(N^3)$  training cost remains unchanged, while the additional overhead comes from evaluating the prior-model transformation. Since the prior-model parameters are jointly learned together with the GP hyperparameters, backpropagation through the prior model also increases the constant-factor cost, but does not change the overall complexity order.

For the Half-Cheetah and 7-DOF Reacher systems, GP-PI-K uses SVGP together with a fixed surrogate neural network as the prior model. In this case, compared with a standard SVGP, the dominant complexity order remains that of the underlying variational GP training and inference, while the additional overhead comes from the forward evaluation of the surrogate neural network and the additive composite kernel. Therefore, in both the exact GP and SVGP settings, GP-PI-K introduces additional computational overhead relative to standard GP models, but does not change the overall complexity class.

In addition to the theoretical complexity analysis, we also empirically compare the runtime of all methods on all four systems. Specifically, we report the wall-clock training time and the wall-clock inference time for one-step prediction on the full test set. The results are summarized in Table 24. Overall, the empirical results are consistent with the theoretical analysis: compared with standard GP, GP-PI-K requires additional training and inference time due to the prior-model transformation and the composite-kernel evaluation. However, this increase mainly reflects additional constant-factor overhead, rather than a change in the overall complexity class.

Table 24 also shows that GP-PI-K is generally more computationally demanding than the other GP-based baselines, which is expected because it combines prior-model transformation with an additive composite kernel. Nevertheless, the empirical trends remain consistent with the theoretical analysis in Table 23.





**Table 23**

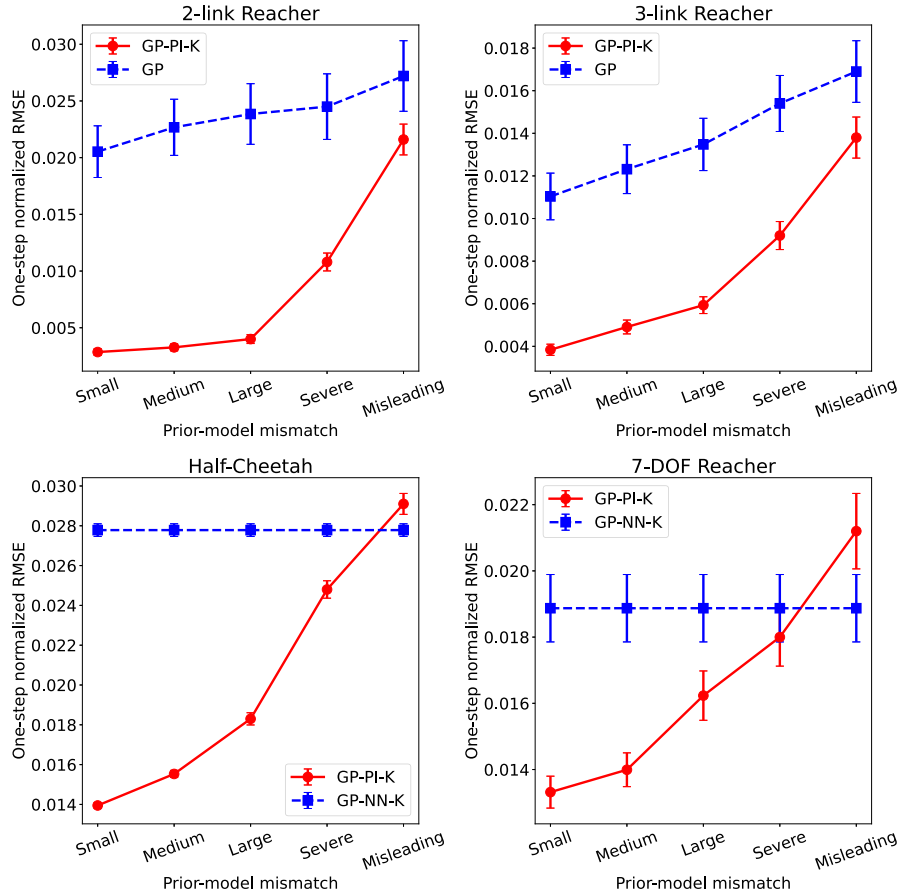
Theoretical computational complexity of standard GP and GP-PI-K. Here,  $N$  denotes the number of training samples,  $M$  the number of test samples,  $B$  the mini-batch size, and  $M_u$  the number of inducing points.  $C_\phi$  denotes the computational cost of one prior-model evaluation.

| Setting  | Model                           | Training complexity                                                | Inference complexity                       |
|----------|---------------------------------|--------------------------------------------------------------------|--------------------------------------------|
| Exact GP | Standard GP                     | $\mathcal{O}(N^3)$                                                 | $\mathcal{O}(MN^2)$                        |
| Exact GP | GP-PI-K with ODE prior          | $\mathcal{O}(N^3) + \mathcal{O}(NC_\phi)$                          | $\mathcal{O}(MN^2) + \mathcal{O}(MC_\phi)$ |
| SVGP     | Standard GP                     | $\mathcal{O}(BM_u^2 + M_u^3)$ per iteration                        | $\mathcal{O}(MM_u)$                        |
| SVGP     | GP-PI-K with surrogate NN prior | $\mathcal{O}(BM_u^2 + M_u^3) + \mathcal{O}(BC_\phi)$ per iteration | $\mathcal{O}(MM_u) + \mathcal{O}(MC_\phi)$ |

**Table 24**

Empirical runtime comparison of different models on all four systems. Training time denotes the wall-clock time for model training, and inference time denotes the wall-clock time for one-step prediction on the full test set. All runtime values are reported in seconds.

| System         | Phase     | GP-PI-K | GP     | GP-PI-M | GP-NN-M | GP-NN-K | DE    |
|----------------|-----------|---------|--------|---------|---------|---------|-------|
| 2-link Reacher | Training  | 48.7    | 10.6   | 11.1    | 10.9    | 15.0    | 4.3   |
|                | Inference | 4.9     | 2.2    | 2.5     | 2.8     | 3.0     | 0.004 |
| 3-link Reacher | Training  | 410.3   | 70.8   | 90.2    | 74.6    | 123.1   | 24.5  |
|                | Inference | 18.5    | 6.7    | 6.0     | 6.9     | 6.8     | 0.41  |
| Half-Cheetah   | Training  | 7254.0  | 1322.5 | 1583.2  | 3134.1  | 2118.1  | 852.3 |
|                | Inference | 39.2    | 13.4   | 18.7    | 19.7    | 15.2    | 10.7  |
| 7-DOF Reacher  | Training  | 1371.4  | 237.8  | 281.7   | 564.4   | 383.4   | 101.8 |
|                | Inference | 20.4    | 12.6   | 14.0    | 14.8    | 15.6    | 10.9  |



**Fig. 12.** One-step normalized RMSE under increasing prior-model mismatch. The prior-model mismatch levels are extended from small, medium, and large to severe and misleading to examine the performance decline and failure cases of GP-PI-K. For the 2-link and 3-link Reacher systems, GP is used as the strongest non-prior baseline. For Half-Cheetah and 7-DOF Reacher, GP-NN-K is used as the strongest non-prior baseline. Error bars show 95% confidence intervals over 20 random seeds.

## 6. Conclusion

This paper proposes a novel kernel design approach in which the input space is transformed using either analytical physical models or imperfect simulators, resulting in a prior-model-informed kernel. This design enhances the extrapolation capability of GPs while largely preserving the favorable uncertainty behavior of standard GPs. Experimental results show that our proposed model consistently outperforms existing GP-based models in terms of both model prediction performance and downstream tasks. The kernel-level integration of prior knowledge proves to be more robust than mean-function-based alternatives.

For future work, an important direction is to develop a fully Bayesian treatment of the physical prior parameters. In the present work, these parameters are optimized under constraints with weak regularization for computational tractability. Future work may treat them as random variables with explicit prior distributions and infer their posterior distribution within a hierarchical Bayesian GP framework. We also plan to investigate hierarchical kernel structures that combine multiple prior models with varying levels of fidelity. Another promising direction is to extend the framework to high-dimensional settings, such as learning latent dynamics from raw image observations in visual control tasks.

### CRedit authorship contribution statement

**Shengbing Tang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Chen Xiao:** Data curation; **Bin He:** Funding acquisition; **Na Li:** Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. Implementation details and reproducibility

### A.1. GP training settings

For the 2-link Reacher and 3-link Reacher systems, we use exact GP inference. For the Half-Cheetah and 7-DOF Reacher systems, we use stochastic variational Gaussian processes (SVGPs), because the training datasets are substantially larger. For all systems, each output dimension is modeled independently.

For the SVGP models, we use a Cholesky variational distribution together with a variational strategy in which the inducing locations are learned during training. The number of inducing points is set to 128 per output dimension. The inducing points are initialized using the first 128 normalized training inputs for each output dimension. We use Gaussian likelihoods and optimize the variational evidence lower bound (ELBO) with Adam. Unless otherwise stated, the learning rate is 0.003, the training mini-batch size is 2000, and the maximum number of training epochs is fixed to 2500. No early stopping is used. For all models, both inputs and outputs are normalized using precomputed dataset statistics. The exact GP and SVGP implementations are based on GPyTorch.

### A.2. Kernel settings

For the 2-link Reacher and 3-link Reacher systems, we use RBF kernels for the GP-based models. In GP-PI-K, the prior-model-informed base kernel is implemented as an RBF kernel applied to the one-dimensional ODE prior-model output for each output dimension, and the residual kernel is implemented as an RBF kernel defined on the original input space with ARD lengthscales. For GP, GP-PI-M, GP-NN-M, and GP-NN-K, the same RBF kernel family is used whenever applicable.

For the Half-Cheetah and 7-DOF Reacher systems, the prior-model-informed base kernel in GP-PI-K is implemented as a Matérn kernel with  $\nu = 2.5$  applied to the one-dimensional surrogate NN prior-model output for each output dimension. The residual kernel is also implemented as a Matérn kernel with  $\nu = 2.5$ , but is defined on the original input space using ARD lengthscales. For GP, GP-PI-M, GP-NN-M, and GP-NN-K, the same Matérn kernel family with  $\nu = 2.5$  is used whenever applicable. This ensures that differences among GP-based methods are not caused by using different kernel families.

### A.3. Neural-network architectures, parameter counts, and compute budgets

For the Half-Cheetah and 7-DOF Reacher systems, the simulator-based prior model used in GP-PI-K is represented by a surrogate neural network. The surrogate network consists of three hidden layers, each with 256 hidden units and ReLU activations, followed by a linear output layer. After pretraining on prior-generated data, its parameters are fixed and are not updated during GP-PI-K training.

For the GP-NN-K baseline, we use a neural network to transform the input before applying the GP kernel. The architecture of this neural feature extractor depends on the system size. For the 2-link Reacher system, the feature extractor has two hidden layers with 64 hidden units per layer and outputs 20-dimensional features. For the 3-link Reacher system, it has two hidden layers with 256 hidden units per layer and outputs 30-dimensional features. For the Half-Cheetah and 7-DOF Reacher systems, it has three hidden layers with 256 hidden units per layer and outputs 50-dimensional features. SiLU activations are used in the hidden layers. The output features are normalized by subtracting the feature-wise mean and dividing by the feature-wise standard deviation within each batch. All output dimensions share the same neural-network feature extractor. The transformed features are then used inside the GP kernel.

For the GP-NN-M baseline, we use a neural-network mean function followed by a GP trained on residuals. The neural network predicts all output dimensions simultaneously. It is first trained using mean squared error with Adam. Afterward, the residual targets are computed by subtracting the neural-network predictions from the normalized outputs, and a GP model is trained on these residuals.

For DE, each ensemble member is implemented as an MLP dynamics model that directly predicts the next state. The ensemble size is fixed to  $M = 10$ . To make the comparison between GP-NN-K and DE fair, we use matched hidden-layer backbones for the two methods on each system. Specifically, GP-NN-K and each DE ensemble member use the same hidden-layer depth, hidden-layer width, activation function, optimizer, learning rate, mini-batch size, and maximum training budget. The only architecture difference is in the output layer: GP-NN-K outputs learned features used inside the GP kernel, whereas each DE member outputs the next-state prediction. For DE, we report both the per-member parameter count and the total parameter count across all ensemble members. The corresponding neural-network parameter counts and training times for GP-NN-K and DE are summarized in [Table A.25](#).

No separate hyperparameter search is performed for GP-NN-K or DE. Instead, fixed hyperparameter settings and matched compute budgets are used across random seeds. Here, compute budget refers to the allowed training resources, including the maximum number of training epochs, mini-batch size, optimizer, learning rate, number of random seeds, ensemble size for DE, number of inducing points for

**Table A.25**

Matched neural-network parameter counts and training times for GP-NN-K and DE. GP-NN-K and each DE ensemble member use matched hidden-layer backbones on each system. Here,  $M$  denotes the ensemble size of DE. For DE, both the per-member parameter count and the total parameter count across all ensemble members are reported. Training time is reported in seconds under the same runtime measurement protocol as Table 24. For DE, the training time refers to the total time for training the full ensemble.

| System         | Method  | NN backbone                        | Params/network | Total NN params | Training time |
|----------------|---------|------------------------------------|----------------|-----------------|---------------|
| 2-link Reacher | GP-NN-K | $2 \times 64$ MLP, SiLU            | 6036           | 6036            | 15.0          |
|                | DE      | $2 \times 64$ MLP, SiLU, $M = 10$  | 5126           | 51,260          | 4.3           |
| 3-link Reacher | GP-NN-K | $2 \times 256$ MLP, SiLU           | 76,830         | 76,830          | 123.1         |
|                | DE      | $2 \times 256$ MLP, SiLU, $M = 10$ | 71,433         | 714,330         | 24.5          |
| Half-Cheetah   | GP-NN-K | $3 \times 256$ MLP, SiLU           | 152,626        | 152,626         | 2118.1        |
|                | DE      | $3 \times 256$ MLP, SiLU, $M = 10$ | 146,201        | 1,462,010       | 852.3         |
| 7-DOF Reacher  | GP-NN-K | $3 \times 256$ MLP, SiLU           | 151,858        | 151,858         | 383.4         |
|                | DE      | $3 \times 256$ MLP, SiLU, $M = 10$ | 144,405        | 1,444,050       | 101.8         |

GP-NN-K, and the common hardware/software environment used for runtime measurement. The same training datasets, test datasets, and 20 random seeds are used for both methods. Runtime is measured under the same protocol as Table 24. Therefore, the comparison between GP-NN-K and DE is based on matched neural-network backbones and fixed training budgets, rather than method-specific hyperparameter tuning.

#### A.4. Data collection protocols

For the 2-link Reacher and 3-link Reacher systems, the training data are collected using random actions.

For the Half-Cheetah and 7-DOF Reacher systems, the training data consist of two parts. The first part is collected using random actions, which improves coverage of the state-action space. The second part is collected from trajectories generated by model-free reinforcement learning policies, which provide high-reward and task-relevant transitions. For the Half-Cheetah system, the model-free policy is obtained using Soft Actor-Critic (SAC). For the 7-DOF Reacher system, the model-free policy is obtained using Trust Region Policy Optimization (TRPO), since SAC does not solve this task reliably. The two datasets are then combined to train the GP-based dynamics models.

#### A.5. Planning setup in MBRL

In model-based reinforcement learning, planning is performed using model predictive control (MPC) combined with the cross-entropy method (CEM). At each environment step, the planner optimizes an action sequence over a finite planning horizon and executes only the first action, after which planning is repeated at the next step. The task horizons used in different systems are summarized in Table 13 in the main text. For all systems, the MPC planning horizon is 25.

For each planning step, the action-sequence distribution is initialized as a Gaussian distribution with zero mean and standard deviation 0.1. CEM then iteratively samples candidate action sequences, evaluates them using the learned dynamics model, selects the elite candidates, and updates the action distribution. Specifically, the number of CEM iterations is 5, the number of sampled action sequences is 2000, the number of elite sequences is 100, and the smoothing coefficient for updating the action distribution is 0.2.

To evaluate a candidate action sequence, we propagate the learned dynamics model forward over the MPC planning horizon and accumulate the predicted rewards. A particle-based evaluation strategy is used, in which each candidate action sequence is evaluated using multiple stochastic rollouts and the mean return over particles is used as its score. The number of particles is 5. During MBRL training, Gaussian exploration noise with standard deviation 0.02 is added to the first action selected by MPC before executing it in the real environment.

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